When and how visual aesthetic features influence approach-avoidance motivated behavior

THIS IS A POST-PRINT
PLEASE CITE AS:

Pombo, M., Corradi, G. B., Elliot, A. J., & Velasco, C. (2025). EXPRESS: When and how visual aesthetic features influence approach-avoidance motivated behavior. *Quarterly Journal of Experimental Psychology*, *0*(ja). https://doi.org/10.1177/17470218251371660

Maria Pombo (maria.pombo@nyu.edu)[1], Guido Corradi (guidocor@gmail.com)[2,3], Andrew J. Elliot (andrew.elliot@rochester.edu)[4], & Carlos Velasco (carlos.velasco@bi.no)[5]

1. Department of Psychology, New York University, New York, NY, U.S.A
2. Department of Statistics and Operations Research, Universidad Complutense de Madrid, Madrid, Spain
3. BEATLES Research Group, University of Balearic Islands. Spain
4. Department of Psychology, University of Rochester, Rochester, NY 14627, USA.
5. Centre for Multisensory Marketing, Department of Marketing, BI Norwegian Business School, Oslo, Norway, Nydalsveien 37, 0484 Oslo, Norway

REGISTERED REPORT

TO SUBMIT TO: QUARTERLY JOURNAL OF EXPERIMENTAL PSYCHOLOGY

Abstract

For decades, researchers have explored the relationship between aesthetic features such as symmetry and complexity and preference. Likewise, philosophers and psychologists alike have pondered the differences between preference and behavior. Nevertheless, little is known about the relationship between aesthetic preference and motivation. Using an online approach-avoidance motivation task, we compare the reaction times between congruent (aesthetic preference + approach) and incongruent (aesthetic preference + avoidance) conditions. In Experiment 1, we explore the relationship between symmetry and complexity and approach-avoidance motivated behavior. In Experiment 2, we distinguish the mechanisms in play in the relationship by manipulating presentation and decision-making time. In Experiment 3, we assess the generalizability of the results beyond abstract graphical patterns. Overall, our research advances knowledge of the relationship between aesthetic preferences for features like symmetry and complexity and approach-avoidance motivation, and it has implications for fields such as marketing.

Keywords: aesthetics, motivation, approach, avoidance, symmetry, complexity

Introduction

The link between aesthetics and motivation is exhibited in tasks as ubiquitous as buying a garment that looks pretty or agreeing to go on a date with someone attractive. Humans appear to be motivated by things they like. However, despite considerable empirical work on aesthetics and the rise of the field of neuroaesthetics, the question of how aesthetics relates to approach-avoidance motivated behavior, remains unresolved (Chatterjee & Vartanian, 2014; Skov & Nadal, 2020).

What the focus of neuroaesthetics should be remains one of the largest controversies in the field (Brielmann, 2021). On one hand, some researchers define neuroaesthetics as the study of the biological bases of aesthetic experiences (Chatterjee & Vartanian, 2014). Based on this definition, there are certain objects, or object properties, that elicit a particular aesthetic response. On the other hand, a more broad account of neuroaesthetics defines it as the study of how sensory information acquires hedonic value (Skov & Nadal, 2020; Wassiliwizky & Menninghaus, 2021).

Both accounts describe the relationship between preference and motivation differently. Researchers arguing for a more narrow approach hypothesize that disinterested interest, a mental state characterized by a deep engagement with an object without a desire to acquire or control it, may resemble neural activity for “liking” but not “wanting”, and may be accompanied by the experience of pleasurable aesthetic emotions (Hayn-Leichsenring & Chatterjee, 2018). If true, this could explain how aesthetic experiences can be byproducts of rewarding experiences. At the same time, aesthetic experiences would then differ from the desire for objects that drives behaviors such as consumer behavior. Conversely, researchers arguing for a broader account

hypothesize that humans compute the hedonic value of sensory information with the functional goal of motivating behavioral interactions with them (Chatterjee & Vartanian, 2014). It is worth noting that, based on this broader account, there are a plethora of studies in fields like food hedonics and neuroeconomics that explore the relationship between hedonic valuation and approach or avoidance motivated behavior (e.g., Havermans, 2011; Kahneman et al., 1993; Piqueras-Fiszman et al., 2014).

Arguing for or against what the focus of neuroaesthetics should be extends beyond this paper. Nevertheless, there are contexts for which knowing how certain features, deemed as aesthetic features by the narrow approach (Palmer et al., 2013), relate to approach or avoidance motivated behavior is highly relevant (Chatterjee & Vartanian, 2014). These include visual art, product design,and marketing. For example, marketers may wonder how visual features (e.g., spatial structure of objects, colors) in their brand elements (e.g., packaging, ads, websites) will lead to specific motivational behaviors. We thus rely on the narrower definition of aesthetics to explore the tie between specific visual aesthetic features and motivation, in particular approach-avoidance motivation.

Preference and Motivation

Previously, researchers have studied the connection between preference and behavior, determining that even though liking and wanting may be separate systems (Berridge et al., 2009), their corresponding neural circuitry partially overlap (Anselme & Robinson, 2016; Castro & Berridge, 2014; Rangel et al., 2008).

Taking this further, there has been some research focusing on understanding the relationship between aesthetics and motivation. For example, Berlyne's theory of aesthetics has been discussed in the field for several decades (Cupchik, 1986; Marin et al., 2016; Menninghaus

et al., 2019; Nadal et al., 2010, p. 201; Skov, 2019). Berlyne's research studied the relationship between both interest and attention and motivational processes. He explored how curiosity, the need for satiation, and the intrinsic elements of objects, such as their complexity, evoke interest (Berlyne, 1949, 1960, 1971, 1974). Overall, he viewed aesthetic activity as a form of exploration rooted in intrinsic motivation (Cupchik, 1986). Though some have found experimental support for his theories (Friedenberg & Liby, 2016; Imamoglu, 2000), others have not (Güçlütürk et al., 2016; Martindale et al., 1990). Furthermore, his theory concerning the connection between arousal and aesthetic appeal has been widely criticized, with claims that it is an oversimplification and not concrete enough to explain how arousal and motivation relate to affective responses (Silvia, 2005; Skov, 2019).

Additional research has established that when the motivation to engage with artwork has been induced in participants, whether by asking them to generate a title or by telling them that engaging with artwork has benefits, they reported feeling significantly more pleasure than when they simply rated the paintings based on their gut feeling (Steciuch et al., 2019). Other studies have shown ties between learning motivation and visual aesthetic ratings (H.-F. Wang, 2019) and that acquiring motivation modulates whether aesthetics impact consumer decisions (L. Wang et al., 2019).

Our focus herein is on approach and avoidance motivated behavior. Approach motivation involves appetition, reward, and incentive, whereas avoidance motivation involves aversion, punishment, and threat (Elliot et al., 2013). Approach and avoidance motivation are thought to be regulated by two primary systems, the Behavioral Activation System (BAS) and the Behavioral Inhibition System (BIS) (Carver & White, 1994; Elliot et al., 2013; Gray, 1982), with the BAS

responsible for approach motivated behavior and the BIS responsible for avoidance motivated behavior.

Some existing work has examined the relationship between aesthetics and approach and avoidance motivated behavior. Vessel and colleagues (2018) looked at shared preferences across aesthetic domains. Their study included a keypress task, in which participants could increase or decrease the viewing time of an image by pressing a key. They argued that keypress viewing time is a measure of approach motivation, and their results suggest that keypress patterns, for the most part, resemble those of preference ratings. Vartanian and colleagues (2013) used a categorization task to study how differences in architecture contours influence aesthetic evaluations ("beautiful" or "not beautiful") and approach-avoidance decisions ("enter" or "exit"). Their results suggest that even though contour influenced aesthetic evaluations (participants judged curvilinear spaces as more beautiful), contour did not affect approach-avoidance decisions (Vartanian et al., 2019). Velasco and colleagues (2020) found that symmetrical (as opposed to asymmetrical) shapes are associated with approach words. Furthermore, in a separate study, Velasco and colleagues (2020) manipulated the symmetry of packaging and the hedonic value of different products and measured their association to approach and avoidance words using a two-alternative forced-choice task. Their results suggest that regardless of the hedonic value of the product, symmetrical packaging designs are more likely to be associated with approach words than asymmetrical packaging designs.

While these studies provide some insight into how people associate aesthetic features and approach and avoidance categories, they are mostly based on categorization tasks and may contain "liking" as a possible confound. In other words, it is not possible to know whether their participants related different aesthetic features with approach and avoidance categories because

of their motivational meaning and not, say, their valence. Thus, two important questions remain: What is the relationship between visual aesthetic stimulus features and approach-avoidance motivated behavior? What explains such a relationship (if, indeed, it exists)? In the present research, we aim to address these questions.

To study the relationship between aesthetics and approach and avoidance motivated behavior, we focus on two aesthetic features: symmetry and complexity. These features have been studied extensively, not only in the context of motivation (Berlyne, 1960, 1971; Velasco et al., 2016, 2020) but also in empirical aesthetics research. The existing literature on symmetry and complexity in the context of aesthetics provides us with a strong foundation for the present research. In addition, exploring two different aesthetic features allows us to extend (i.e., generalize) our results.

Symmetry

Although many different types of symmetry exist (e.g., translational, rotational), reflectional or bilateral symmetry is the most studied and the most relevant to humans (Treder, 2010; Turoman et al., 2018). In this case, we define a shape as symmetrical if it is composed of a reflection along a vertical or horizontal midline. Symmetry is a commonly preferred aesthetic feature (Bertamini & Rampone, 2020; Palmer et al., 2013) and preference for symmetry is present across cultures, testing contexts, and stimulus types (Tinio & Leder, 2009). However, preference may be modulated by art expertise (Leder et al., 2019; Weichselbaum et al., 2018). Several theories have tried to explain why symmetry is a preferred aesthetic feature, such as processing fluency theory (Reber et al., 2004), which posits that the brain processes symmetry fluently, or the mere-exposure effect (Zajonc, 1968), which posits that symmetry is a preferred feature given its ubiquity in nature (Treder, 2010). Alternatively, other researchers have

proposed that symmetry is preferred due to its characteristic as an adaptive phenotype (Møller & Thornhill, 1998) or its ability to facilitate image segmentation (Treder, 2010). While the relationship between symmetry and preference is strong, the relationship between symmetry and motivation is not clear. Symmetry's association to quality (Little, 2014; Møller & Thornhill, 1998; Pombo & Velasco, 2021) and the idea that individuals have a tendency to approach elements with high quality (Velasco et al., 2020) would suggest why symmetry (vs. asymmetry) results in approach behavior. This is consistent with the proposition that visual features that regulate approach and avoidance behaviors may also evoke specific aesthetic judgements (Vartanian et al., 2013).

Complexity

The relationship between complexity and preference is not as clear as that between symmetry and preference. Though defining complexity is difficult and definitions may differ across fields, in aesthetics complexity generally refers to the amount of information an element conveys. It can refer to quantity, but it can also be defined in comparative terms (e.g., X is more complex than Y) (Jacobsen & Höfel, 2001). Berlyne (1971) suggested that preference for complexity resembles an inverted U-shape, where too little or too much complexity is not preferred. Some research suggests that too much complexity may violate the natural order, and an intermediate amount of complexity may be preferred because it balances maximizing information and maintaining comprehensibility (Aleem et al., 2017). Alternatively, other researchers have observed a linear relationship between pleasure and complexity. Stamps (2002) suggested that the relationship between pleasure and entropy is either linear or asymptotic, and Nadal and colleagues (2010) suggested that there are different types of complexity and each relates to beauty ratings differently. For example, they proposed that there is a positive linear

relationship between the variety of elements and preference, but an inverted U-shape relationship between asymmetry and preference.

Overall, both symmetry and complexity are generally preferred in a variety of contexts and constitute two of the objectively defined characteristics of beauty (Aleem et al., 2019). Given that these two aesthetic features have been clearly tied to preference and, more tentatively, to motivation (whether theoretically or in categorization tasks), we use them herein as means to assess the relationship between aesthetics and approach and avoidance behavior. In other words, we are assuming the preference for both symmetry and complexity (as extensively supported by previous work), to assess whether these aesthetic features are tied to approach or avoidance motivated behavior.

We hypothesize that more preferred aesthetic features will be tied to approach motivated behavior (as opposed to avoidance motivated behavior). This would be consistent with previous work showing this relationship categorically (Velasco et al., 2016). If our hypothesized relationship were to be true, it would challenge the assumption that visual aesthetic features are characterized by a disinterested interest mental state.

Possible Mechanisms Underlying The Tie between Aesthetics and Motivation

In addition to measuring approach-avoidance behavior in relation to two aesthetic features, the present research aims to begin to understand the possible mechanisms that may explain such relationships. Previous research has established that both presentation time and time pressure when making a decision influence people's evaluation of aesthetic stimuli. Based on the idea that the human brain processes information received at different moments in time differently (Caplette et al., 2020), the effects of presentation time and time pressure may help explain the circumstances under which aesthetic features influence motivated behavior.

Presentation Time

On one hand, following Bar and Neta's results (2006, 2007), Corradi and colleagues (2019) showed that presentation time modulates preference for curvature: curved real-life objects are preferred more at shorter presentation times, and curved meaningless patterns are preferred more at unlimited presentation times. They suggest that in absence of semantic associations, contour is more relevant to the decision, which could explain the preference for curvature only in real object pictures shown at short presentation times. Additionally, they showed that when contour is the only key factor in a decision, as in the abstract patterns without time restriction, contour weighs more on the decision. Their tentative conclusion is that contour elicits a more bottom-up automatic positive response which may be occluded by top-down semantic associations. This aligns with previous research suggesting not only that low-level visual elements require less information and less time to process compared to semantic information, but also that visual preferences depend on the semantic content of stimuli (Vessel et al., 2018; Vessel & Rubin, 2010). Overall, presentation time serves to operationalize potentially different mechanisms for evaluating aesthetic elements. Therefore, it could be the case that shortened presentation times of valenced stimuli may, in a bottom-up fashion, lead to different motivational reactions, whereas longer presentation times allow further access to semantic memory, and thus result in top-down control of the motivated behavior.

Time Pressure

On the other hand, time pressure influences the mechanisms through which aesthetic features are processed. Several theories, both within and outside of aesthetics, suggest an interplay between automatic and voluntary processes in decision making (Kahneman, 2011; Dhar & Gorlin, 2013; Graf & Landwehr, 2015; Leder et al., 2004; Leder & Nadal, 2014). In other

words, preference ratings may vary depending on whether they were made through deliberate or intuitive processing. Thus, these theories would predict that aesthetic-based decisions would differ under induced time pressure.

In the context of consumer behavior, the role of time pressure on decision making has interested researchers for decades (Bettman, 1979; Iyer, 1989). More specifically, some have investigated how this time pressure relates to how aesthetic features are attended to and perceived at the time of making a decision. One study found that under time pressure, consumers filter textual information more (e.g., ingredient information on a shampoo bottle) and pictorial information less (e.g., a logo or picture on a shampoo bottle) (Pieters & Warlop, 1999). Similarly, contours and shapes, as opposed to design features such as size, brand letters, brand pictures, and color, attract consumer's initial visual attention (Clement et al., 2013). Visual elements (compared to informational elements) play a larger role in the decision-making process under time-pressure (Reutskaja et al., 2011). Generally, these results suggest that under time pressure, different aesthetic elements affect attention and perception differently. Overall, the dual-processing theory of decision making, in conjunction with the idea that aesthetic features may be perceived and processed differently under time pressure, highlight the importance of considering time pressure as a modulator of how aesthetic features relate to approach-avoidance motivated behavior.

The ways in which presentation time and decision-making time result in motivational change could provide insight into the processing stage at which aesthetic features affect approach or avoidance motivated behavior. Through three experiments involving an online approach-avoidance task, we examine the relationship between visual aesthetic features and approach-avoidance motivated behavior and determine whether presentation time and decision-making

time may explain this relationship. With such empirical work, we aim to begin to elucidate the relationship between aesthetics and motivated behavior, as well as the mechanisms that may explain this relationship. We predict not only that aesthetic features are related to approach-avoidance motivated behavior in both abstract and consumer contexts, but also that presentation time and time pressure modulate this relationship. Table 1 provides the details of our research questions, hypotheses, and data collection and analysis plans.

Hypotheses and Research Plan

Table 1. Design Table

Question	Hypotheses	Sampling Plan	Analysis Plan	Interpretation given to different outcomes
What is the relationship between visual aesthetic features and approach-avoidance behavior?	For both symmetry and complexity conditions: **$H1_0$:** Aesthetic features have no influence on approach-avoidance behavior: there is no significant difference in reaction time between congruent and incongruent, correct trials. **$H1_1$:** Aesthetic features influence approach-avoidance behavior: reaction times of congruent correct trials are significantly faster than reaction times of incongruent, correct trials. Note: congruent trials are when symmetry or high complexity is paired with approach and incongruent trials are when symmetry or high complexity is paired with avoidance.	Based on a simulation-based power calculation (described in detail below) we will recruit 190 participants for the symmetry condition and 130 participants for the complexity condition.	We will use linear mixed effects models (one for each aesthetic variable) to calculate the effect of congruency on reaction times. We will have a random intercept for both participants and stimuli. We will calculate the significance of our beta estimates with Kenward-Roger approximation (a = 0.05) and report how well our model accounts for the variability of our data (pseudo-R-squared).	If $p < 0.05$ for the estimate of the congruency fixed effect, then we can reject the null hypothesis and conclude we have evidence to suggest that the average RT for congruent trials is significantly different from the average RT for incongruent trials. If the reaction time for incongruent trials is larger, that would suggest an association between symmetry or high complexity and approach motivated behavior, and asymmetry or low complexity and avoidance behavior. If $p > 0.05$, we fail to reject H0 and conclude we do not have evidence of an association between aesthetic features and approach-avoidance motivated behavior.

Does the stage of cognitive processing of the aesthetic features determine the relationship between the aesthetic features and approach-avoidance behavior?	**$H2_0$:** The interaction effect of presentation time and congruency on reaction time is not significant. **$H2_1$:** At shorter presentation times, the effect of aesthetic features on approach-avoidance motivated behavior is stronger: at short presentation times, the difference between the reaction time of congruent correct trials and the reaction time of incongruent correct trials is greater than at long presentation times. **$H3_0$:** The interaction effect of decision time and congruency on reaction time is not significant. **$H3_1$**: When participants must decide in a short amount of time, the effect of aesthetic features on approach-avoidance motivated behavior is stronger: when decision time is short, the difference between the reaction time of congruent correct trials and the reaction time of incongruent correct trials is greater than at long presentation times. **$H4_0$:** The three-way interaction effect between presentation times, decision time, and congruency is not significant. **$H4_1$:** At shorter presentation time the difference in reaction time between the congruent and incongruent trials for the immediate decision time is significantly	We will recruit the same number of participants as in the experiment above: 190 for the symmetry condition and 130 for the complexity condition.	We will use two linear mixed effect models (one for each aesthetic feature) with random intercepts for participants and stimuli. We will include congruency (congruent vs. incongruent), presentation time (fast vs. slow), and decision time (immediately vs. delayed) as fixed effects. We will also include the interaction between congruency and presentation time, congruency and decision time, and the triple interaction of congruency, presentation time, and decision time. We will calculate the significance of our beta estimates (a = 0.05) and report how well our model accounts for the variability of our data (pseudo-R-squared). Lastly, if the interactions are significant, we will conduct pairwise comparisons to figure out the directionality of the effects. We will correct	If $p < 0.05$ for the estimate of the congruency fixed effect, then we can reject the null hypothesis and conclude we have evidence to suggest that all else being equal, the average RT for congruent trials is significantly different from the average RT for incongruent trials. If the reaction time for incongruent trials is larger, that would suggest an association between symmetry or high complexity and approach motivated behavior, and asymmetry or low complexity and avoidance behavior. If $p < 0.05$ for the estimate of the presentation time effect, then we can reject the null hypothesis and conclude we have evidence to suggest that all else being equal, the average RT for trials with short presentation time is significantly different from the average RT for trials with long presentation time, suggesting an association between presentation time and reaction time. If $p < 0.05$ for the estimate of the decision time effect, then we can reject the null hypothesis and conclude we have evidence to suggest that all else being equal, the

	different than the difference in reaction time between the congruent and incongruent trials for the delayed decision time.	the *p*-values for multiple comparisons.	average RT for trials with immediate decision time significantly differs from the average RT for trials with delayed decision time, suggesting an association between decision time and reaction time. If $p < 0.05$ for the estimate of any of the interactions, we will conclude that we have evidence to suggest a significant interaction between the variables. Using the pairwise comparisons and using $p < 0.05$ as a guideline, we will determine the conditions in which reaction times are significantly different. If $p > 0.05$, we fail to reject H0 and conclude we do not have evidence of an association between aesthetic features and presentation time, decision time, or approach-avoidance motivated behavior.
Do the observed relationships generalize to consumer contexts?	Here we will attempt to replicate all previous experiments using more ecologically valid stimuli that resemble products encountered by consumers. Our hypotheses, sampling, and analysis plans will be the same.		In the case that we fully replicate our findings, we will conclude that aesthetic features in consumer products behave similarly to aesthetic features in abstract shapes. In the case that we do not (or

	partially) replicate our findings, we will discuss potential reasons for the observed differences.

Methods

Ethics Information

This study was approved by the ethics committee at the Universidad Camilo José Cela (CEI-UCJC, with code 14_CEI_2020) and was carried out in accordance with the World Medical Association's Declaration of Helsinki. We obtained informed consent from all participants and participants were compensated in accordance with Prolific Academic's (https://prolific.com) participant payment guidelines.

Experiment 1

Participants

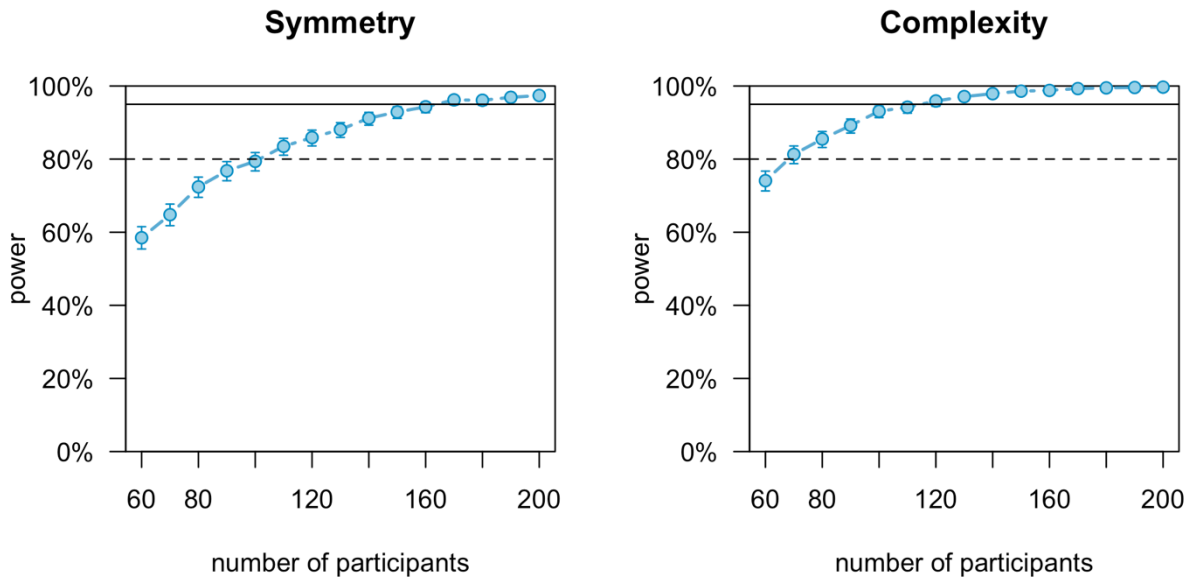
190 participants took part in the symmetry experiment (110 female, 80 male; ages 19-72, $M = 42.2$, $SD = 14.3$). 128 took part in the complexity experiment (65 female, 62 male, 1 other; ages 19-74, $M = 40.2$, $SD = 13.9$). It took participants, on average, approximately 8 minutes to complete the study.

Sampling Plan

There is no satisfactory analytical solution to calculate the power of relatively complex linear mixed effects models, so we relied on a simulation-based power estimation (Kumle et al., 2021). The simulations were conducted in R using the "simr" package (Green & MacLeod, 2016). We estimated how many times we would get a statistically significant result if we were to run our experiment 1000 times as a function of number of participants. We used the parameter estimates from our pilot study. Figure 1 shows the results of our simulations. Based on these

simulations, we selected the sample size based on 95% power (shown as a bold horizontal line in the plots) using the lower limit of the confidence intervals. For the symmetry condition, we recruited 190 participants and for the complexity condition we recruited 130 participants. We recruited participants that indicated that they are based in the U.K., speak English as a first language, and have normal or corrected-to-normal vision.

Fig 1.

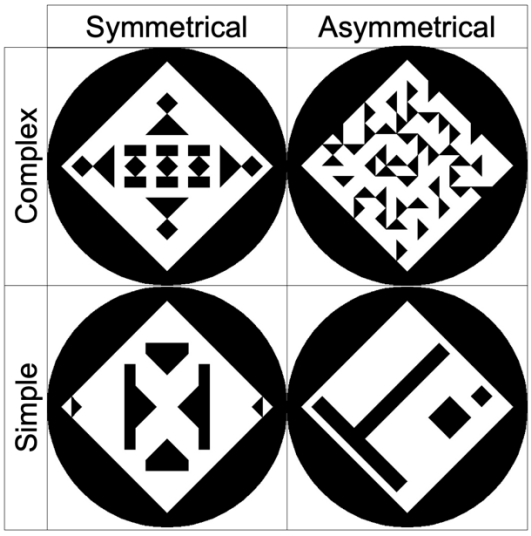


Stimuli

We obtained permission to use the graphic pattern stimuli from Jacobesen and Höfel (2002) in our study. We decided to use these stimuli because they have been widely used in aesthetic experiments (Augustin et al., 2012; Jacobsen, 2010; Jacobsen et al., 2006; Leder et al., 2019). These stimuli (252 total) vary in terms of symmetry (number of symmetry axes) and complexity (number of elements). The symmetry condition contains both symmetrical and asymmetrical stimuli. For the symmetrical stimuli, we selected all of the stimuli that were

exclusively symmetrical along both the vertical and horizontal axes (60 total). For the asymmetrical stimuli, we randomly selected 60 of the stimuli that contained no axes of symmetry. The mean number of elements in both the symmetrical and asymmetrical pool of stimuli are very similar (14.8 vs. 14.7 respectively). The complexity condition contained both low complexity and high complexity stimuli. To obtain these, we selected the stimuli with the highest number of elements and with the lowest number of elements in each of the symmetry levels. For example, if 20 of the 252 (8%) stimuli are symmetrical along one axis, then 4% of the stimuli in the high-complexity pool are symmetrical along one axis and 4% of the stimuli in the low-complexity pool are symmetrical along one axis. Each pool of complexity stimuli (high-complexity vs. low-complexity) contains 60 stimuli. See Figure 2 for an example of the stimuli used.

Fig 2.



Apparatus and Materials

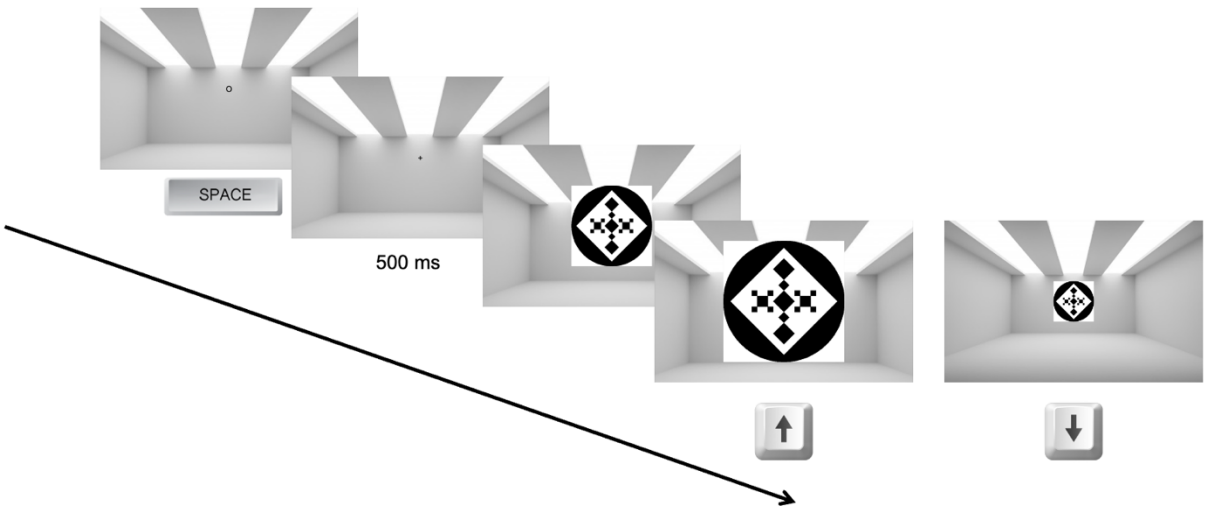
All experiments were programmed using the builder view of PsychoPy3 (Peirce et al., 2019) and delivered through Pavlovia (https://pavlovia.org). The experiment only ran if participants were using a computer, as opposed to a tablet or smartphone. It is worth noting that online experiments, even ones relying on reaction times and presentation times, can provide reliable results (Bridges et al., 2020; Woods et al., 2015).

Design and Procedure

A variety of approach-avoidance tasks (AAT) exist to measure approach-avoidance motivated behavior. The original AAT (Solarz, 1960), as well as AATs requiring specialized equipment, such as the Joystick AAT (Rinck & Becker, 2007) or mouse tracking and mouse pushing and pulling tasks (Weil et al., 2017; Wittekind et al., 2015), are appropriate for laboratory settings. However, AATs have been adapted for online studies resulting in other options: the Manikin AAT (Markman & Brendl, 2005), the Mobile AAT (Zech et al., 2020), the Tilt task (Kakoschke et al., 2018), and the Swiping task (Meule et al., 2019). One particular task, the online VAAST task (Aubé et al., 2019), appears to be promising to measure approach-avoidance behavior in online settings. Many of the alternative online tasks use arm movements, which can be ambiguous given that arm extension and flexion can signal both approach and avoidance. The VAAST task simulates whole body movement without requiring actual body movement and results suggest large and replicable compatibility effects (Rougier et al., 2018). Likewise, the online VAAST replicates the results of the VAAST task when conducted in laboratory settings and results are consistent across cultures (Aubé et al., 2019). Overall, the online VAAST task provides an easy and ecologically valid way to measure approach-avoidance behavior in an online study and thus was our chosen task. .

Our online VAAST task had a mixed design with aesthetic feature (symmetry vs. complexity) as a between-participant condition and congruency of trial (congruent vs. incongruent) as a within-participant condition. Each participant took part in either the symmetry or the complexity online VAAST. Participants gave informed consent and then were told that they should use the upwards and downwards arrow on their computer to move forward and backwards in a given environment, in this case a white room. Participants then completed two tasks. In one task, which we refer to as the congruent condition, they were told that they had to approach, by pressing the upwards arrow on their keyboard, symmetrical (complex) shapes and avoid, by pressing the downwards arrow on their keyboard, asymmetrical (simple) shapes. The instructions gave participants a definition of symmetry (complexity). Participants were asked to respond as fast and as correctly as possible and were asked to only use the index finger of their dominant hand. The task contained 6 practice trials and 40 experimental trials. There were 20 trials for each aesthetic condition (e.g., 20 symmetrical shapes and 20 asymmetrical shapes) and for each participant, these were randomly selected out of the total pool of 60 stimuli. During practice trials, which we excludedfrom the analysis, participants received feedback on whether they were correct (e.g., pressed the upwards arrow upon the appearance of a symmetrical stimulus). They received no feedback on the remaining trials. Participants had to press the spacebar to start each trial. A fixation cross appeared for half a second and then the shape appeared. Each time the participant pressed the upwards arrow, the stimulus got bigger, and every time they pressed the downwards arrow the stimulus got smaller, simulating approach behavior and avoidance behavior, respectively. We measured participants' reaction time to press an arrow upon presentation of the stimulus, as well as whether they got it correct. The second task was the same, except we asked participants to do the opposite (e.g., approach asymmetrical

shapes and avoid symmetrical shapes). The second task, which we refer to as the incongruent condition, used the same stimuli as the first task, and the order in which participants completed the tasks was counterbalanced. Note that the second task also included 6 practice trials that were excluded from the analysis. Figure 3 shows an example of what a single trial looked like. Lastly, we asked participants whether they noticed the changes in the instructions and to rate, on a Likert-type scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree), how much they agree with the following statement: "During the entire experiment, I was paying attention and responded to the best of my abilities".



Analysis Plan

We excluded participants who did not notice the change in instructions, who rated their attention on the task a 3 (Neither agree nor disagree) or below, and who replied incorrectly to more than 20% of the trials in both congruency conditions. First, we considered the trials in which participants answered correctly and for which the reaction time fell within the mean reaction time ± two standard deviations.

For each type of aesthetic feature (symmetry vs. complexity), we ran a linear mixed model in R using the "lmer" function in the "lme4" package (Bates et al., 2015). Unlike traditional ANOVAs, mixed-effects models allow accounting for both between and within-participant factors (DeBruine & Barr, 2021; Judd et al., 2017). In particular, mixed-effects models allow us to account for effects of both participants and stimuli in addition to the effect of congruency on reaction times. In our model, we included random intercept values for both participants and stimuli. We also included congruency level (congruent vs. incongruent trials) as a fixed effect. Since reaction times tend to follow a non-normal distribution, we took their logarithm to adjust the residuals. We had separate models for participants in the symmetry and complexity conditions. We used the Kenward-Roger approximation of the "lmerTest" package to obtain *p*-values and we assessed the effectiveness of our model with the conditional pseudoR-squared value provided by the "model_performance()" function in R (Lüdecke et al., 2020; Luke, 2017).

To complement our reaction time analysis, we also considered whether there is a difference in the number of incorrect responses between congruent and incongruent trials using a generalized linear mixed model.

Pilot Data

To test the feasibility of Experiment 1, and since the following experiments assume that there is a relationship between aesthetic features and approach and avoidance motivated behavior, we conducted a pilot experiment (N = 200) following the design of Experiment 1. We found that participants were, on average, faster at approaching symmetrical stimuli and avoiding asymmetrical stimuli, and slower at avoiding symmetrical stimuli and approaching asymmetrical stimuli, $t(6424) = 2.65$, $b = 0.02$, $p < 0.008$, 95% CI = [0.01, 0.04]. We also found that

participants were, on average, faster at approaching complex stimuli and avoiding simple stimuli, and slower at avoiding complex stimuli and approaching simple stimuli, $t(5824) = 3.58$, $b = 0.02$, $p < 0.001$, 95% CI = [0.01, 0.02]. For a full description of the pilot analyses, see the Supplementary Information. All data and analyses are available at: https://osf.io/pa9df.

.

Experiment 2

Participants

184 participants took part in the symmetry experiment (79 female, 104 male, 1 other; ages 19-77, $M = 41.1$, $SD = 12.7$). 127 took part in the complexity experiment (54 female, 71 male, 2 other; ages 19-77, $M = 39.2$, $SD = 13.0$). It took participants, on average, approximately 27 minutes to complete the study.

Stimuli, Apparatus, and Materials

For Experiment 2 we used the same pools of symmetry and complexity stimuli as in Experiment 1. We used the same method to develop and distribute Experiment 2 as used in Experiment 1.

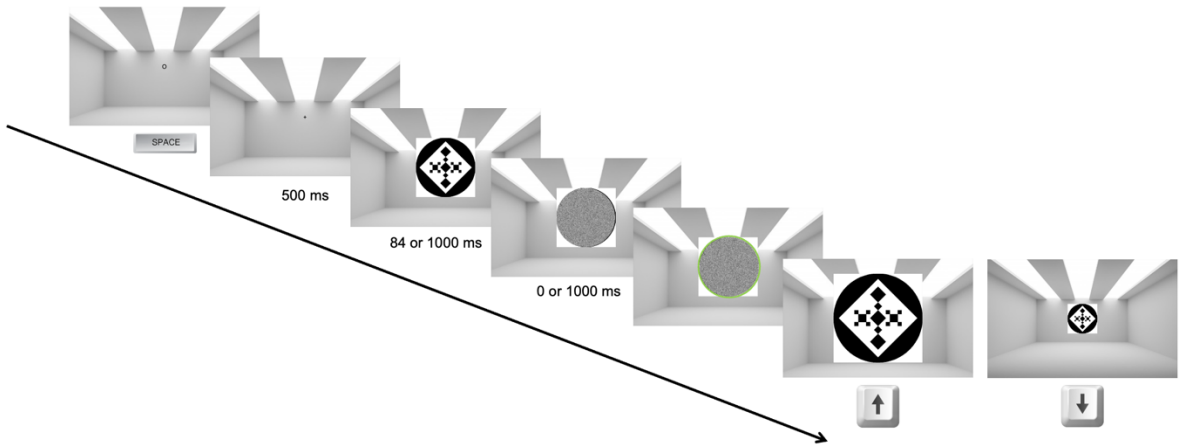
Design and Procedure

Experiment 2 consisted of an online VAAST task. This experiment followed a mixed design with aesthetic features (symmetry vs. complexity) as a between-participant variable, and congruency of trial (congruent vs. incongruent), presentation time (fast vs. slow), and time to make a decision (immediately vs. delayed) as within participant variables. We considered stimuli presented for 84 milliseconds as “fast”. We chose 84 milliseconds in accordance with findings

that suggest that participants more strongly prefer curvature when the stimulus is presented for that duration (Bar & Neta, 2006; Corradi et al., 2019). Additionally, we considered stimuli presented for 1000 milliseconds as "slow". Even though 1000 milliseconds seems fast when comparing the ecological validity of aesthetic judgments (e.g., one may stare at a painting for several minutes), previous research has shown that there are no differences in aesthetic judgments between presentation times of 1 second and 30 seconds (Brielmann et al., 2017), and individuals only need 750 milliseconds to make an aesthetic judgment of a musical piece (Belfi et al., 2018). Participants were asked to either make a decision (i.e., press a key in response to the VAAST task) immediately after stimulus offset or after 1000 milliseconds. It was predicted that the mean choice reaction time for young adults for a visual stimulus is around 350 milliseconds (Solanki et al., 2012). Thus, if participants were asked to respond after 1000 milliseconds, they would have enough time to avoid an impulse reaction and change their mind.

The design of Experiment 2 differed slightly from Experiment 1. Since the images had a strict presentation time, they were covered by a mask after the appropriate amount of time. To indicate that participants were allowed to respond, a green circle appeared around the mask. Figure 4 shows the experimental procedure. To facilitate the task for participants, the decision and presentation time conditions were separated by blocks. Participants completed 8 blocks, one for each combination of congruency, presentation time, and decision-making time. Like Experiment 1, each participant only completed the experiment for one aesthetic feature. The order in which participants completed each block was randomized.

Fig 4.



Sampling Plan

We recruited participants in the same way and following the same criteria as in Experiment 1.

Analysis Plan

We used two linear mixed effect models, one for each aesthetic feature, with random intercepts for participants and stimuli. We included congruency (congruent vs. incongruent), presentation time (fast vs. slow), and decision time (immediately vs. delayed) as fixed effects. We also included the interaction between congruency and presentation time, congruency and decision time, and the triple interaction of congruency, presentation time, and decision time. We calculated the significance of our beta estimates (a = 0.05) and reported how well our model accounts for the variability of our data (pseudoR-squared). Lastly, if the interactions were significant, we conducted pairwise comparisons to determine the directionality of the effects and we corrected the p-values for multiple comparisons.

Experiment 3

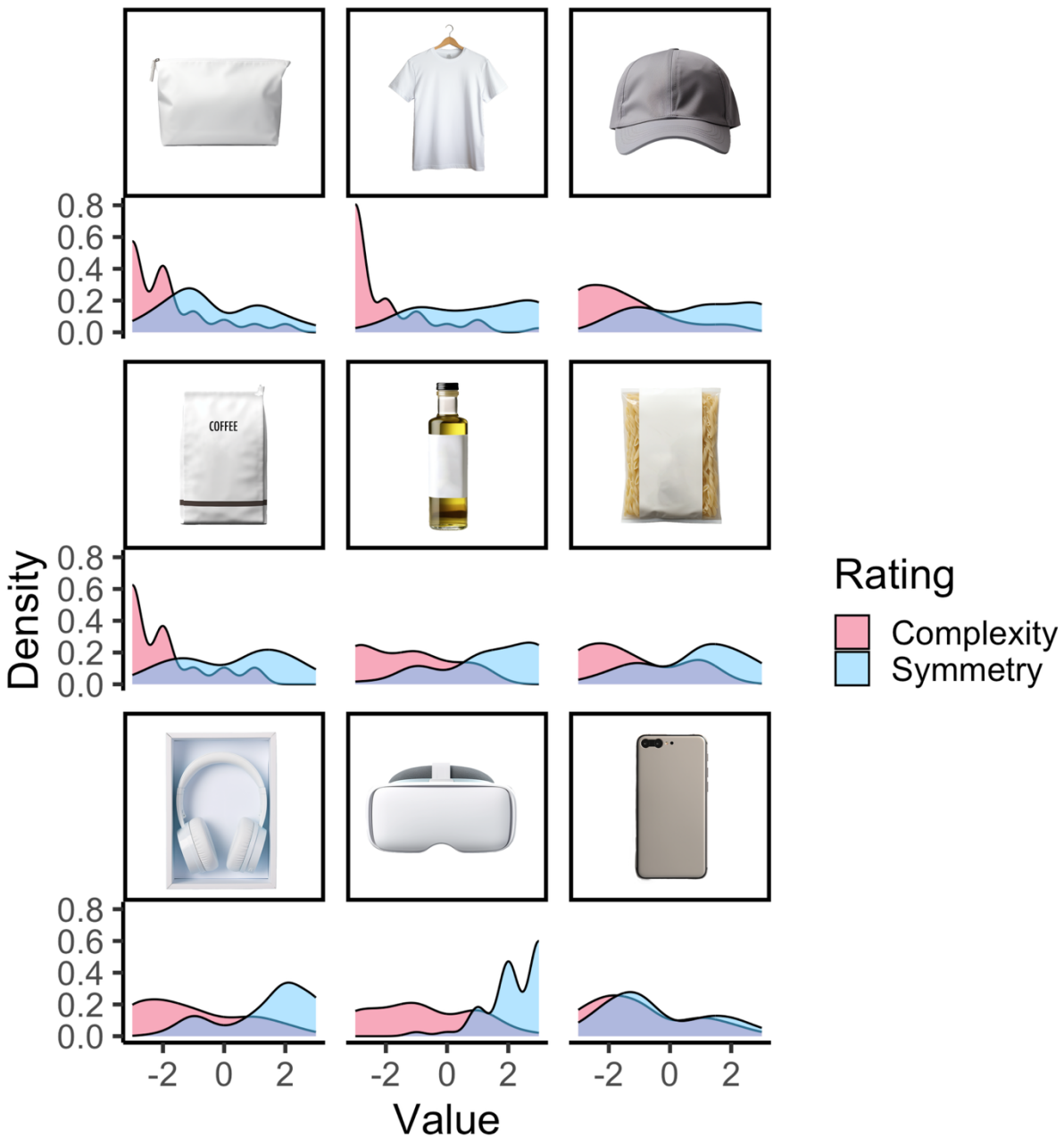
Participants

185 participants took part in the symmetry experiment (95 female, 89 male, 1 other; ages 18-77, $M = 35.9$, $SD = 13.3$). 123 took part in the complexity experiment (58 female, 64 male, 1 other; ages 18-73, $M = 34.5$, $SD = 12.5$). It took participants, on average, approximately 27 minutes to complete the study.

Stimuli

For the third experiment, we created our stimuli by overlaying the symmetry and complexity stimuli from Experiments 1 and 2 onto commonplace consumer goods. We considered three types of product categories: fashion, food packaging, and technology. We used the categorization of each of the overlaid graphical patterns from Experiments 1 and 2 to determine each stimulus' characterization as symmetrical (high complexity) or asymmetrical (low complexity) in accordance with Experiments 1 and 2. To prevent the symmetry or complexity of the consumer goods from interacting with the overlaid pattern, we ran a short pilot study in which we asked participants to rate how symmetrical (complex) the consumer good designs (without the overlaid pattern) was. We tried three exemplars of each category: fashion (t-shirt, hat, bag), food packaging (coffee bag, pasta bag, olive oil bottle), and technology (phone case, headphones, VR headset). Based on the participant ratings, we selected the stimuli with the most uniform distributions of symmetry and complexity ratings. Figure 5 shows the exemplars and the distribution of their symmetry and complexity ratings. Raw data is available in the supplementary material.

Fig 5.



For exploratory purposes, we included the average symmetry or complexity value given to each stimulus as part of our linear mixed effects models to test whether these modulate any of our results.

Design, Sampling, and Analysis Plan

Experiment 3 aimed to replicate Experiment 2 with a different, more ecologically valid, set of stimuli. The design, procedure, sampling plan, and analysis plan were similar to Experiment 2.

We feared that triplicating Experiment 2 to include all three consumer products made the experiment too long (approx. 27 vs. 81 min), resulting in participant fatigue effects. Therefore, we decided to maintain the same length as Experiment 2, keeping the 40 trials per condition, but randomizing the products across trials. As such, participants saw on average 13 trials of each product in each condition. To maintain ecological validity, we presented the three products at different sizes, in accordance to the product's size in real life. Figure 6 shows examples of the three products with an overlaid stimulus.

Fig 6.



Data Availability

Our raw data and materials are available here: https://osf.io/pa9df.

Code Availability

All code used for our analyses is available here: https://osf.io/pa9df.

Results

Experiment 1

Based on our exclusion criteria outlined above, we excluded participants with self-reported attention below a 3 out of 5 and who reported not having noticed the change in instructions between the congruent and incongruent blocks (24 for the symmetry experiment and 15 for the complexity one). We also excluded participants who got more than 20% of the trials incorrect (12 for symmetry, 15 for complexity). We also excluded incorrect trials and trials with response times outside two standard deviations (we allowed $0 < RT < 4.89$ s for symmetry and $0.46 < RT < 2.24$ s for complexity). Below, we report the results of 153 participants (12,059 trials) for the symmetry condition and 98 (7,536 trials) for the complexity condition. Though the resulting sample sizes are below the target size calculated to achieve 95% power, based on our original simulations (Figure 1), our power remains over 90%.

The results of our linear models suggest a significant effect of congruency on reaction times for both symmetry (Table 2) and complexity (Table 3). We took the congruent condition as the reference level. For the symmetry condition, the mean reaction time in the congruent condition was 1.62 seconds ($SD = 0.61$), while the mean reaction time in the incongruent condition was 1.75 seconds ($SD = 0.66$). In other words, participants were, on average, faster at approaching symmetrical stimuli and avoiding asymmetrical stimuli, and slower at avoiding symmetrical stimuli and approaching asymmetrical stimuli. For the complexity condition, the mean reaction time in the congruent condition was 1.26 seconds ($SD = 0.25$) while the mean

reaction time in the incongruent condition was 1.28 seconds (*SD* = 0.26). In other words, participants were, on average, faster at approaching complex stimuli and avoiding simple stimuli, and slower at avoiding complex stimuli and approaching simple stimuli. Thus, we found evidence to support **H1**$_1$. Figure 7 provides a graphical representation of the data.

Fig 7.

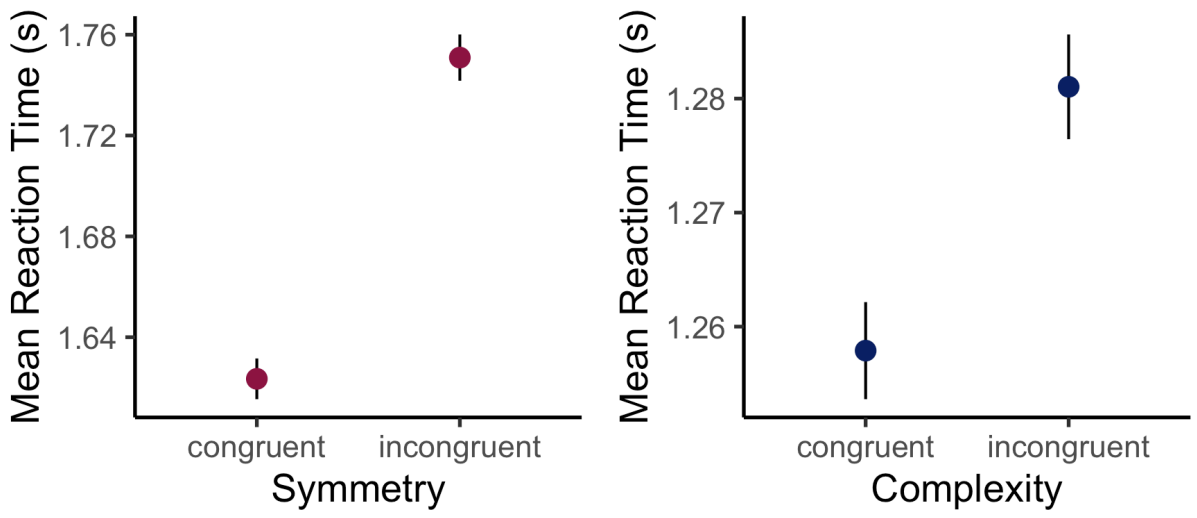


Table 2. Mixed-effects model for the symmetry condition in Experiment 1.

Random Effects

	Variance	***SD***
Stimulus	0.00	0.07
Participant	0.05	0.22

Fixed Effects

	Estimate	***SE***	**95% CI**	***df***	***t***	***p***
Intercept	0.43	0.02	[-0.20, 0.03]	197.312	23.75	<0.001
Congruency	0.07	0.00	[0.19, 0.25]	10578.52	15.51	<0.001

This model explains 48% of the conditional variance and 1% of the marginal variance in log reaction times with RMSE of 0.23. ICC = 0.47.

Table 3. Mixed-effects model for the complexity condition in Experiment 1

Random Effects

	Variance	***SD***
Stimulus	0.00	0.03
Participant	0.01	0.11

Fixed Effects

	Estimate	***SE***	**95% CI**	***df***	***t***	***p***
Intercept	0.22	0.01	[-0.12, 0.13]	111.86	18.47	<0.001
Congruency	0.02	0.00	[0.04, 0.12]	6308.80	4.03	<0.001

This model explains 37% of the conditional variance and 0.2% of the marginal variance in log reaction times with RMSE of 0.15. ICC = 0.37.

Experiment 2

For the results of Experiment 2, we excluded participants with self-reported attention below a 3 out of 5 and who reported not having noticed the change in instructions between the congruent and incongruent blocks (32 for the symmetry experiment and 19 for the complexity one). We also excluded participants who got more than 20% of the trials incorrect in both conditions (13 for symmetry, 20 for complexity). We also excluded incorrect trials and trials with reaction times outside two standard deviations (we allowed $0 < RT < 3.82$ s for symmetry and $0 < RT < 4.4$ s for complexity). Below, we report the results of 139 participants (44,267 trials) for the symmetry condition and 88 (28,027 trials) for the complexity condition. Though the resulting sample sizes are below the target size calculated to achieve 95% power, based on our original simulations (Figure 1), our power remains over 90%.

In our linear models, we took the congruent condition as the reference level for congruency. The reference levels for presentation time and response delay were short presentation (0.084 s) and no delay (0 s) respectively. For symmetry, the proposed linear model did not converge, so instead we display the results of a simpler model, with only participants as a random intercept (Table 4). This model shows a significant interaction between congruency and presentation time, but not one between congruency and response delay. We also found a significant three-way interaction. Figure 8A shows these results. More specifically, we find that there is a larger difference in reaction time for congruent trials ($M = 0.49$ s, $SD = 0.42$) and incongruent trials ($M = 0.52$ s, $SD = 0.42$) during 0.084 s presentation times, than there is during 1 s presentation times ($M_{congruent} = 0.49$ s, $SD = 0.38$; $M_{incongruent} = 0.50$ s, $SD = 0.41$). Thus, we rejected $\mathbf{H2_0}$ and found evidence in support of $\mathbf{H2_1}$. It is worth noting that even though the

interaction between congruency and response delay was not significant, the results do show a tendency towards a larger difference in reaction time without a response delay compared to a 1 s response delay. However, since we did not find a significant interaction between congruency and response delay, we failed to reject $\mathbf{H3_0}$. We found that at short presentation times, the mean difference in reaction time is greater without a response delay ($M_{congruent}$= 0.48 s, SD = 0.39; $M_{incongruent}$= 0.52 s, SD = 0.42), than with one ($M_{congruent}$= 0.51 s, SD = 0.46; $M_{incongruent}$= 0.53 s, SD = 0.43). Thus, we rejected $\mathbf{H4_0}$ and found evidence in support of $\mathbf{H4_1}$.

Fig 8.

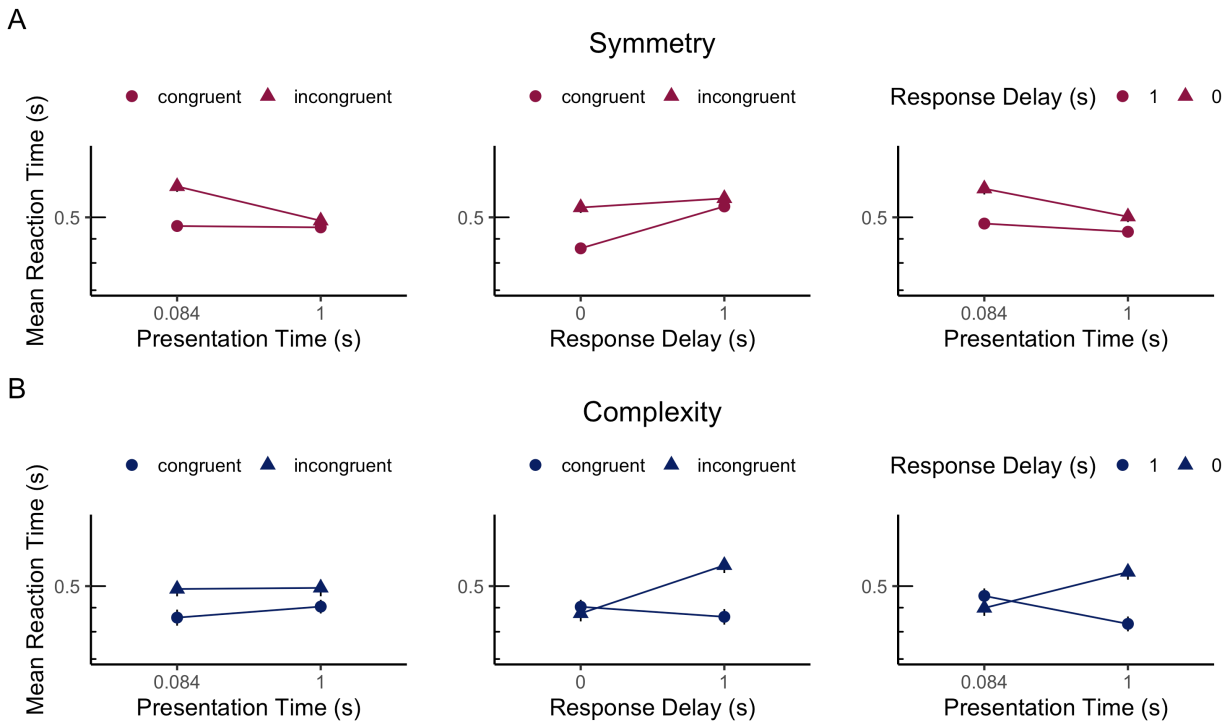


Table 4. Mixed-effects model for the symmetry condition in Experiment 2.

Random Effects

	Variance	***SD***
Participant	0.23	0.46

Fixed Effects

	Estimate	***SE***	**95% CI**	***df***	***t***	***p***
Intercept	-1.09	0.04	[-0.10, 0.05]	174.67	-26.10	<0.001
Congruency	0.03	0.02	[-0.01, 0.07]	40280.86	1.43	0.15
Presentation Time	-0.07	0.02	[-0.10, -0.02]	40380.88	-3.15	0.002
Response Delay	0.05	0.02	[0.00, 0.08]	40371.77	2.18	0.03
Congruency × Presentation Time	0.11	0.03	[0.04, 0.16]	40204.29	3.11	0.002
Congruency × Response Delay	-0.04	0.03	[-0.09, 0.03]	40141.29	-1.01	0.29
Presentation Time × Response Delay	0.10	0.03	[0.04, 0.15]	40381.78	3.35	<0.001
Congruency × Presentation Time × Response Delay	-0.22	0.04	[-0.28, -0.11]	40178.04	4.48	<0.001

This model explains 17.6% of the conditional variance and 0.2% of the marginal variance in log reaction times with RMSE of 1. ICC = 0.18.

For complexity, we found significant interactions between congruency and presentation time and congruency and response delay (Table 5). Here, we did not observe a three-way interaction. The difference in reaction times between congruent and incongruent trials was larger for a 0.084 s presentation time ($M_{congruent}$= 0.48 s, SD = 0.41; $M_{incongruent}$= 0.50 s, SD = 0.43) than for a 1 s presentation time ($M_{congruent}$= 0.49 s, SD = 0.40; $M_{incongruent}$= 0.50 s, SD = 0.45). Thus,

we rejected **H2$_0$** and found evidence in support of **H2$_1$**. The difference in reaction times between congruent and incongruent trials was also larger with a response delay ($M_{congruent}$= 0.48 s, *SD* = 0.41; $M_{incongruent}$= 0.51 s, *SD* = 0.45) than without a delay ($M_{congruent}$= 0.49 s, *SD* = 0.40; $M_{incongruent}$= 0.48 s, *SD* = 0.43). Thus, we rejected **H3$_0$** and found evidence to support the opposite of **H3$_1$**. Rather than observing a larger difference without a response delay, we observe that the effect of congruency is stronger with a 1 s response delay. Since we did not observe a three-way interaction, we failed to reject **H4$_0$**.

Table 5. Mixed-effects model for the complexity condition in Experiment 2.

Random Effects

	Variance	***SD***
Stimulus	0.00	0.03
Participant	0.16	0.40

Fixed Effects

	Estimate	***SE***	**95% CI**	***df***	***t***	***p***
Intercept	-1.06	0.05	[-0.07, 0.11]	135.83	-21.88	<0.001
Congruency	0.00	0.03	[-0.05, 0.06]	23326.56	0.07	0.94
Presentation Time	0.04	0.03	[-0.01, 0.09]	23464.79	1.53	0.12
Response Delay	-0.16	0.03	[-0.20, -0.09]	23442.04	-4.98	<0.001
Congruency × Presentation Time	-0.19	0.05	[-0.25 -0.08]	23156.44	-3.96	<0.001
Congruency × Response Delay	0.19	0.05	[0.09, 0.26]	23185.54	4.16	<0.001
Presentation Time × Response Delay	0.10	0.04	[0.02, 0.17]	23445.53	2.42	0.02
Congruency × Presentation Time × Response Delay	0.03	0.06	[-0.09, 0.14]	23155.26	0.48	0.63

This model explains 14.1% of the conditional variance and 0.4% of the marginal variance in log reaction times with RMSE of 1.01. ICC = 0.137.

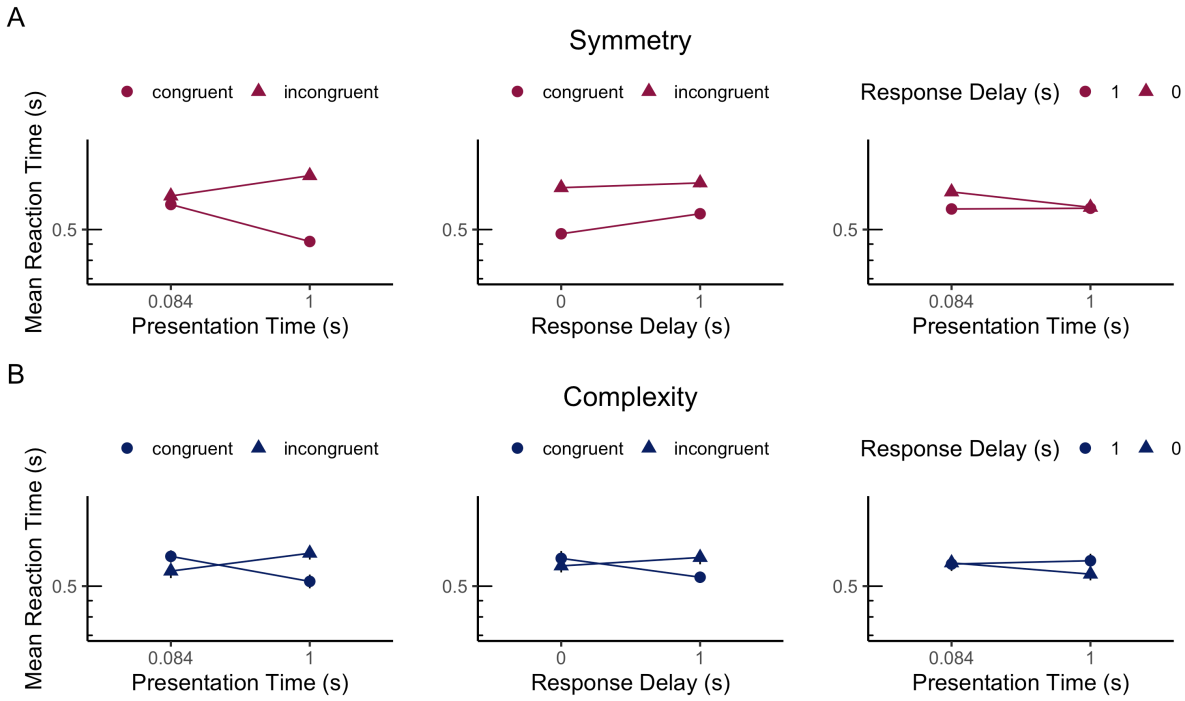
Experiment 3

For the results of Experiment 3, we excluded participants with self-reported attention below a 3 out of 5 and who reported not having noticed the change in instructions between the congruent and incongruent blocks (32 for the symmetry experiment and 26 for the complexity one). We also excluded participants who got more than 20% of the trials incorrect in both conditions (33 for symmetry, 39 for complexity). We also excluded incorrect trials and trials with reaction times outside two standard deviations (we allowed $0 < RT < 4.87$ s for symmetry and $0 < RT < 4.4$ s for complexity). Below, we report the results of 120 participants (33,617 trials) for the symmetry condition and 58 (15,314 trials) for the complexity condition. Though the resulting sample sizes are below the target size calculated to achieve 95% power, based on our original simulations (Figure 1), our power remains over 85% for symmetry and 75% for complexity.

Our proposed mixed-effects linear models did not converge. Thus, we display the results for a simpler, alternative model that only includes participants as a random intercept. We took the congruent condition as the reference level for congruency. The reference levels for presentation time and response delay were short presentation (0.084 s) and no delay (0 s) respectively.

For symmetry, our results indicate significant interactions between congruency and presentation time and congruency and response delay (Table 6). We also observed a significant three-way interaction. More specifically, there was a larger difference in reaction time between congruent and incongruent trials during 1 s presentation times ($M_{congruent}$= 0.49 s, SD = 0.40; $M_{incongruent}$= 0.56 s, SD = 0.48) than during 0.084 s presentation times ($M_{congruent}$= 0.53 s, SD = 0.45; $M_{incongruent}$= 0.54 s, SD = 0.47). Thus, we rejected $\mathbf{H2_0}$, but found evidence to suggest the

opposite of **H2$_1$**. The effect of congruency on reaction time was larger at longer presentation times. Regarding response delay, there was a larger difference between reaction times for congruent and incongruent trials without a response delay ($M_{congruent}$= 0.50 s, SD = 0.41; $M_{incongruent}$= 0.55 s, SD = 0.47) than with a 1 s delay ($M_{congruent}$= 0.52 s, SD = 0.44; $M_{incongruent}$= 0.56 s, SD = 0.48). Thus, we rejected **H3$_0$** and found evidence to support **H3$_1$**. Contrary to our expectation, we found that at short presentation times, the mean difference in reaction time was greater with a 1s response delay ($M_{congruent}$= 0.53 s, SD = 0.45; $M_{incongruent}$= 0.56 s, SD = 0.50), than without ($M_{congruent}$= 0.53 s, SD = 0.45; $M_{incongruent}$= 0.52 s, SD = 0.44). Thus, we rejected **H4$_0$** and found evidence in support of the opposite of **H4$_1$**. Figure 9A shows the results.



********** [INSERT TABLE 6 HERE] *********

For complexity, our model indicates significant interactions between congruency and presentation time and congruency and delay (Table 7). We also observed a significant three-way interaction. More specifically, there is a larger difference in reaction time between congruent and incongruent trials at 1 s presentation times ($M_{congruent}$= 0.51 s, SD = 0.48; $M_{incongruent}$= 0.54 s, SD = 0.45) than at shorter presentation times ($M_{congruent}$= 0.53 s, SD = 0.45; $M_{incongruent}$= 0.52 s, SD = 0.48). Likewise, there is a larger difference in reaction time with a 1 s delay ($M_{congruent}$= 0.51 s, SD = 0.42; $M_{incongruent}$= 0.53 s, SD = 0.46) than without a delay ($M_{congruent}$= 0.53 s, SD = 0.51; $M_{incongruent}$= 0.52 s, SD = 0.47) . Thus, we rejected $\mathbf{H2_0}$ and $\mathbf{H3_0}$, but we find evidence to support the opposite of $\mathbf{H2_1}$ and $\mathbf{H3_1}$: the effect of congruency on reaction time was greater at longer presentation times and without a response delay. Contrary to our expectations, we found that at short presentation times, the mean difference in reaction time was greater with a 1 s response delay ($M_{congruent}$= 0.56 s, SD = 0.44; $M_{incongruent}$= 0.50 s, SD = 0.44), than without ($M_{congruent}$= 0.51 s, SD = 0.45; $M_{incongruent}$= 0.54 s, SD = 0.52). Thus, we rejected $\mathbf{H4_0}$, but found evidence to support the opposite of $\mathbf{H4_1}$. Figure 9B shows the results.

Table 7. Mixed-effects model for the complexity condition in Experiment 3.

Random Effects

	Variance	***SD***
Participant	0.12	0.35

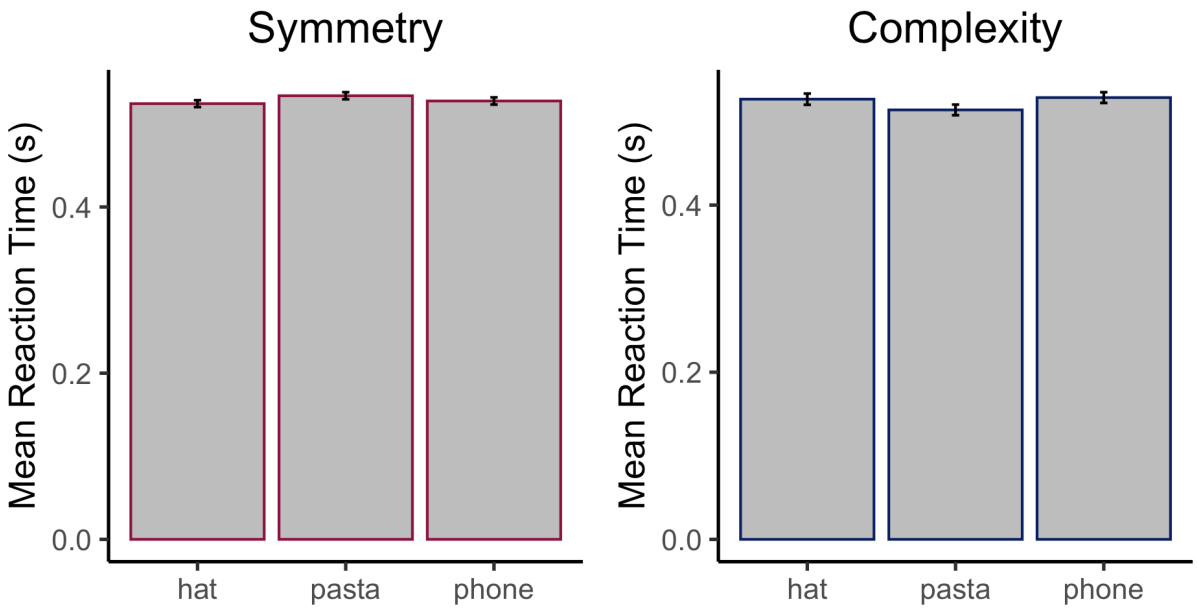
Fixed Effects

	Estimate	***SE***	**95% CI**	***df***	***t***	***p***
Intercept	-1.03	0.05	[-0.15, 0.05]	81.22	-20.35	<0.001
Congruency	0.10	0.04	[0.03, 0.17]	15057.49	2.72	<0.001
Presentation Time	0.15	0.04	[0.08, 0.23]	15109.82	4.21	<0.001
Response Delay	0.11	0.03	[0.04, 0.17]	15144.81	3.17	<0.001
Congruency × Presentation Time	-0.24	0.05	[-0.35, -0.13]	14724.90	-4.37	<0.001
Congruency × Response Delay	-0.19	0.05	[-0.30, -0.09]	14708.70	-3.53	<0.001
Presentation Time × Response Delay	-0.24	0.05	[-0.34, -0.14]	14982.34	-4.76	<0.001
Congruency × Presentation Time × Response Delay	0.40	0.08	[0.25, 0.56]	14316.66	4.97	<0.001

This model explains 12.7% of the conditional variance and 0.03% of the marginal variance in log reaction times with RMSE of 0.92. ICC = 0.13.

To test whether there were differences in reaction time across products, we ran a one-way ANOVA with log reaction time as the dependent variable and product (3 levels: hat, pasta, and phone) as the independent variable. We did not observe differences in reaction time across product conditions for either symmetry, $F(2, 357) = 0.14$, $p = 0.87$, or complexity, $F(2, 171) = 0.19$, $p = 0.83$ (see Figure 10).

Fog 10.



For exploratory purposes, we repeated the same mixed-effects model analysis to test whether the mean symmetry and complexity ratings (Experiment 3 Pilot) modulated any of our effects. We included symmetry or complexity product mean rating as a main effect. The reported main effects and interactions between congruency, presentation time, and response delay time remained, but we did not find any significant effects of either variable on log reaction times.

Discussion

To explore the relationship between visual aesthetic features, specifically symmetry and complexity, we conducted three online experiments using an online approach-avoidance

motivation task. We hypothesized that more preferred aesthetic features (symmetry vs. asymmetry and high complexity vs. low complexity) would elicit approach behaviors rather than avoidance, in both abstract and consumer contexts. By manipulating presentation time and time pressure to make a decision (response delay), we examined the automatic and deliberate processing of aesthetic features, thereby revealing the conditions under which these features strongly affect approach or avoidance motivated behavior.

Experiment 1 showed that participants were generally faster at approaching symmetrical and complex stimuli and slower at approaching asymmetrical and simple stimuli, indicating a main effect of congruency on reaction time. In accordance with our hypotheses, Experiment 2 showed that, for symmetry and complexity, the effect of congruency on reaction time was more pronounced at shorter presentation times. We also observed that for symmetry the effect of congruency is more pronounced without a response delay during short presentation times. However, contrary to our expectations, we only observed an effect of response delay for complexity stimuli, and the effect of congruency was greater with a response delay. Experiment 3, which targets a more ecologically valid context, showed that for symmetry and complexity, the effect of congruency is greater at long presentation times. While the effect of congruency on reaction time was greater without a response delay for symmetry, the opposite was true for complexity. Lastly, we found a significant three-way interaction between congruency, presentation time, and response delay. Table 8 summarizes the results in relation to our hypotheses.

********** [INSERT TABLE 8 HERE] *********

The results of Experiment 1 align with our hypothesis and previous research (Velasco et al., 2016, 2020) suggesting that preferred aesthetic features are linked to approach-motivated behavior. Specifically, aesthetic features like symmetry and complexity, which are generally preferred and have been tied to positive hedonic value (Bertamini & Rampone, 2020; Palmer et al., 2013; Stamps, 2002), directly facilitate approach behaviors. The overlap in neural circuitry between "liking" and "wanting" systems (Anselme & Robinson, 2016; Berridge et al., 2009; Castro & Berridge, 2014; Rangel et al., 2008) may explain this effect, supporting the broader account of neuroaesthetics that posits sensory information acquires hedonic value to motivate interaction (Skov & Nadal, 2020; Wassiliwizky & Menninghaus, 2021). Within the confines of our stimuli and experimental design, these findings may challenge the notion of aesthetic experience as a state of disinterested interest characterized by engagement without desire (Hayn-Leichsenring & Chatterjee, 2018), highlighting instead that aesthetic preferences can actively drive motivational behavior.

The effects found in Experiment 1 are more nuanced when considered in the context of different processing and decision times. Indeed, the findings from Experiments 2 and 3 offer conceptual insights into how presentation time and response delay modulate the relationship between aesthetic features and approach-avoidance motivated behavior. Regarding presentation time, the results of Experiment 2 suggest that symmetrical and complex stimuli elicit immediate, bottom-up responses that rapidly influence approach behaviors (Corradi et al., 2019; Stamps, 2002). This is in accordance with our original hypothesis, and suggests that for abstract patterns, aesthetic features more strongly motivate approach behavior in the absence of top-down semantic associations (Tawil et al., 2024).

Experiment 3 suggests the opposite. Overlaying the same patterns on consumer goods resulted in a stronger effect of congruency for longer presentation times. These results imply that in real-world settings, additional time may be required to process aesthetic features, as well as the semantic information about objects and the categories they belong to, and translate them into motivated action (Nadal et al., 2010; Pombo & Velasco, 2021). This aligns with the perspective that objects with rich affordances demand more cognitive processing (Djebbara & Kalantari, 2023), emphasizing the crucial role of affordances in our interaction with complex stimuli (Gibson, 2014). Importantly, these findings underscore the limitations of assuming that results generalize—claims about aesthetic features in abstract patterns may not consistently extend to more ecologically valid environments. This highlights the importance of conducting studies in more naturalistic settings and capitalizing on modern technologies to develop stimuli with greater ecological validity.

When it comes to response delay, the results are less conclusive. In terms of complexity, for both abstract shapes (Experiment 2) and consumer goods (Experiment 3), the effect of congruency on reaction time was stronger after a 1 second response delay. For symmetry, the effect of congruency on reaction time was stronger without a response delay for consumer goods (Experiment 3). The same tendency was present for abstract patterns, but the interaction between congruency and response delay was not significant. Overall, the interaction between congruency and response time seems to be more driven by aesthetic features themselves, rather than by context (abstract vs. consumer products). A possible reason why symmetry and complexity are differently affected by response delay may be ease of categorization. Even though both symmetry and complexity are operationalized in a plethora of ways (Bertamini & Rampone, 2020; Nadal et al., 2010; Pombo et al., 2023; Treder, 2010), reflection along the vertical and

horizontal axes may be a more prototypical representation of symmetry than the number of elements is of complexity. Thus, it could be that, in our experiment, making a decision about symmetry was easier than making one about complexity. This idea is supported by the larger number of incorrect trials in the complexity experiments compared to the symmetry ones.

Researchers have also shown competition between symmetry and complexity: symmetrical stimuli tend to be less complex (Aleem et al., 2017, 2019; Mather et al., 2023). Individuals thus tend to weigh each of these differently when making an aesthetic judgment. It is plausible that at a population level, one feature is generally weighted more highly than the other, suggesting different interactions between preference and the time required to make a decision. Lastly, the inconclusive nature of the response delay results may also be due to a limitation of our experimental design. Even though we asked participants to wait one second to give a *response*, we could not control the time it took them to make a *decision* about the stimulus. They could make a decision soon after stimulus offset, independent of how long they had to wait to input their decision.

The results of our experiments may be elegantly explained by the affective gradient hypothesis, which suggests that individuals continuously evaluate potential future states based solely on their affective responses (Shenhav, 2024). This theory posits that individuals contemplate all potential future states and the affective response associated with each potential action. These form a gradient that directs behavior towards more desirable and away from less desirable outcomes. Following this hypothesis, symmetrical and complex visual features likely generate stronger positive affective responses, creating a steeper affective gradient that naturally drives individuals to approach these stimuli more readily. Conversely, asymmetrical and simple features may evoke weaker or even negative affect, resulting in lower approach behaviors as

individuals are less motivated to engage with them. These gradients may be influenced both by the time-related processing, as well as semantic information, further modulating said behavior.

In cognitive psychology, reaction time differences as small as 20 milliseconds are often considered meaningful, particularly in tasks that measure implicit biases or automatic cognitive processes. For example, in the study by Rougier et al. (2020), a reaction time difference of 28 milliseconds between compatible and incompatible conditions was found to be significant and relevant in the context of approach-avoidance task, highlighting the sensitivity of such tasks to small temporal differences. This aligns with findings from meta-analyses on evaluative conditioning and implicit association tasks, where differences of 20-30 milliseconds have been shown to reliably indicate underlying cognitive and affective processes (Greenwald et al., 2009; Hofmann et al., 2010). Such small differences are robust indicators of automatic evaluations and behavioral tendencies, even though they may seem trivial in magnitude.

Independent of the mechanisms driving our results, understanding the relationship between aesthetic features and approach/avoidance motivated behavior across contexts has practical implications for domains such as marketing and design, where leveraging aesthetic features can strategically influence consumer evaluations and decision-making (Althuizen, 2021; Bettels & Wiedmann, 2019). Indeed, from a practical standpoint, our findings emphasize the value of integrating preferred aesthetic features like symmetry and complexity into product design, packaging, and other marketing materials to actively facilitate approach behaviors from consumers. In a marketplace where the majority of product launches appear to fail, capitalizing on aesthetic features in design can create a point of differentiation, helping products stand out in a crowded and competitive environment while capturing consumer interest and motivation (Schneider & Hall, 2011). By leveraging these aesthetic elements, marketers can enhance the

attractiveness of their offerings (as long as the promise of the product holds), thereby increasing consumer engagement and the likelihood of purchase decisions (Velasco et al., 2020; H.-F. Wang, 2019). Understanding how aesthetic features interact with presentation time and response delay also provides insights that marketing strategies may capitalize on. Products and their corresponding marketing elements are presented in different formats (e.g., at the shop, e-commerce, in ads, or Youtube videos) that vary in terms of exposure and decision times. The interplay between aesthetic features and said times may be strategically used to maximize impact.

It is important to note that, in the present research, we do not claim that symmetry and complexity are synonymous with aesthetic appeal; rather, our approach builds upon a substantial body of literature indicating that, on average, these visual properties tend to influence aesthetic preferences positively (e,g., Bertamini & Rampone, 2020; Nadal et al., 2010; Palmer et al., 2013; Stamps, 2002; Tinio & Leder, 2009; Treder, 2010; Turoman et al., 2018). However, we acknowledge important exceptions noted in the literature, highlighting evidence that simpler rather than complex patterns may often be perceived as more aesthetically appealing (e.g., McDougall & Reppa, 2008; Reppa & McDougall, 2022), and that symmetry is not always the strongest predictor of perceived beauty (Jones & Jaeger, 2019). In our study, the use of mixed models was intended to account for individual variability in responses to these visual attributes. Nevertheless, future research should explicitly investigate how individual differences in preferences for symmetry and complexity moderate the influence of these attributes on approach or avoidance behaviors. Such studies might benefit from preliminary rating experiments to empirically validate the aesthetic appeal of stimuli before their use, ensuring that the chosen visual features effectively reflect the relevant aesthetic dimensions.

Here, we offer a paradigm for future researchers to expand beyond symmetry and complexity and consider how other features, like curvature, texture, or color, influence approach/avoidance motivated behavior. It is worth considering that while our research moves beyond previous studies assessing approach and avoidance motivation in a way that may be confounded with liking, our design may still involve liking as a potential contributor to our results. Future research may still utilize other approach-avoidance experimental paradigms and give consideration to individual differences in preference to replicate and extend our results.

Conclusion

In conclusion, our findings affirm that visual aesthetic features like symmetry and complexity are linked to approach-avoidance motivated behavior. Preferred aesthetic elements not only elicit approach behaviors but do so through mechanisms influenced by both automatic and deliberate cognitive processing, as modulated by presentation time and response delay. These results underscore the importance of understanding how specific aesthetic features drive approach-avoidance motivation, with practical implications for fields like marketing and product design, where these elements can significantly influence consumer behavior.

References

Aleem, H., Correa-Herran, I., & Grzywacz, N. M. (2017). Inferring Master Painters' Esthetic Biases from the Statistics of Portraits. *Frontiers in Human Neuroscience*, *11*. https://doi.org/10.3389/fnhum.2017.00094

Aleem, H., Pombo, M., Correa-Herran, I., & Grzywacz, N. M. (2019). Is Beauty in the Eye of the Beholder or an Objective Truth? A Neuroscientific Answer. In J. L. Contreras-Vidal, D. Robleto, J. G. Cruz-Garza, J. M. Azorín, & C. S. Nam (Eds.), *Mobile Brain-Body Imaging and the Neuroscience of Art, Innovation and Creativity* (pp. 101–110). Springer International Publishing. https://doi.org/10.1007/978-3-030-24326-5_11

Althuizen, N. (2021). Revisiting Berlyne's inverted U-shape relationship between complexity and liking: The role of effort, arousal, and status in the appreciation of product design aesthetics. *Psychology & Marketing*, *38*(3), 481–503. https://doi.org/10.1002/mar.21449

Anselme, P., & Robinson, M. J. F. (2016). "Wanting," "liking," and their relation to consciousness. *Journal of Experimental Psychology: Animal Learning and Cognition*, *42*(2), 123–140. https://doi.org/10.1037/xan0000090

Aubé, B., Rougier, M., Muller, D., Ric, F., & Yzerbyt, V. (2019). The online-VAAST: A short and online tool to measure spontaneous approach and avoidance tendencies. *Acta Psychologica*, *201*, 102942. https://doi.org/10.1016/j.actpsy.2019.102942

Augustin, M. D., Wagemans, J., & Carbon, C.-C. (2012). All is beautiful? Generality vs. specificity of word usage in visual aesthetics. *Acta Psychologica*, *139*(1), 187–201. https://doi.org/10.1016/j.actpsy.2011.10.004

Bar, M., & Neta, M. (2006). Humans Prefer Curved Visual Objects. *Psychological Science*,

17(8), 645–648. https://doi.org/10.1111/j.1467-9280.2006.01759.x

Bar, M., & Neta, M. (2007). Visual elements of subjective preference modulate amygdala activation. *Neuropsychologia*, *45*(10), 2191–2200. https://doi.org/10.1016/j.neuropsychologia.2007.03.008

Bates, D., Maechler, M., Bolker, B., & Walker, S. (2015). *lme4: Linear mixed-effects models using Eigen and S4.* (Version 1.1-7. 2014) [R Package].

Belfi, A. M., Kasdan, A., Rowland, J., Vessel, E. A., Starr, G. G., & Poeppel, D. (2018). Rapid timing of musical aesthetic judgments. *Journal of Experimental Psychology: General*, *147*(10), 1531. https://doi.org/10.1037/xge0000474

Berlyne, D. E. (1949). 'Interest' as a Psychological Concept. *British Journal of Psychology. General Section*, *39*(4), 184–195. https://doi.org/10.1111/j.2044-8295.1949.tb00219.x

Berlyne, D. E. (1960). *Conflict, arousal, and curiosity* (pp. xii, 350). McGraw-Hill Book Company. https://doi.org/10.1037/11164-000

Berlyne, D. E. (1971). Aesthetics and Psychobiology. *Journal of Aesthetics and Art Criticism*, *31*(4). https://philpapers.org/rec/BERAAP-4

Berlyne, D. E. (1974). *Studies in the new experimental aesthetics: Steps toward an objective psychology of aesthetic appreciation* (pp. viii, 340). Hemisphere.

Berridge, K. C., Robinson, T. E., & Aldridge, J. W. (2009). Dissecting components of reward: 'Liking', 'wanting', and learning. *Current Opinion in Pharmacology*, *9*(1), 65–73. https://doi.org/10.1016/j.coph.2008.12.014

Bertamini, M., & Rampone, G. (2020). The Study of Symmetry in Empirical Aesthetics. In M. Nadal & O. Vartanian (Eds.), *The Oxford Handbook of Empirical Aesthetics*. Oxford University Press. https://doi.org/10.1093/oxfordhb/9780198824350.013.23

Bettels, J., & Wiedmann, K.-P. (2019). Brand logo symmetry and product design: The spillover effects on consumer inferences. *Journal of Business Research*, *97*, 1–9. https://doi.org/10.1016/j.jbusres.2018.12.039

Bettman, J. R. (1979). Memory Factors in Consumer Choice: A Review. *Journal of Marketing*, *43*(2), 37–53. https://doi.org/10.1177/002224297904300205

Bridges, D., Pitiot, A., MacAskill, M. R., & Peirce, J. W. (2020). The timing mega-study: Comparing a range of experiment generators, both lab-based and online. *PeerJ*, *8*, e9414. https://doi.org/10.7717/peerj.9414

Brielmann, A. A. (2021). *Aesthetics, Empirical | Internet Encyclopedia of Philosophy*. https://iep.utm.edu/emp-aest/

Brielmann, A. A., Vale, L., & Pelli, D. G. (2017). Beauty at a glance: The feeling of beauty and the amplitude of pleasure are independent of stimulus duration. *Journal of Vision*, *17*(14), 9–9. https://doi.org/10.1167/17.14.9

Caplette, L., Ince, R. A. A., Jerbi, K., & Gosselin, F. (2020). Disentangling presentation and processing times in the brain. *NeuroImage*, *218*, 116994. https://doi.org/10.1016/j.neuroimage.2020.116994

Carver, C. S., & White, T. L. (1994). Behavioral inhibition, behavioral activation, and affective responses to impeding reward and punishment: The BIS/BAS Scales. *Journal of Personality and Social Psychology*, *67*, 319–333.

Castro, D. C., & Berridge, K. C. (2014, September). *Advances in the neurobiological bases for food "liking" versus "wanting."* Physiology & Behavior; Physiol Behav. https://doi.org/10.1016/j.physbeh.2014.05.022

Chatterjee, A., & Vartanian, O. (2014). Neuroaesthetics. *Trends in Cognitive Sciences*, *18*(7),

370–375. https://doi.org/10.1016/j.tics.2014.03.003

Clement, J., Kristensen, T., & Grønhaug, K. (2013). Understanding consumers' in-store visual perception: The influence of package design features on visual attention. *Journal of Retailing and Consumer Services*, *20*(2), 234–239. https://doi.org/10.1016/j.jretconser.2013.01.003

Corradi, G., Rosselló-Mir, J., Vañó, J., Chuquichambi, E., Bertamini, M., & Munar, E. (2019). The effects of presentation time on preference for curvature of real objects and meaningless novel patterns. *British Journal of Psychology*, *110*(4), 670–685. https://doi.org/10.1111/bjop.12367

Cupchik, G. C. (1986). A decade after Berlyne: New directions in experimental aesthetics. *Poetics*, *15*(4), 345–369. https://doi.org/10.1016/0304-422X(86)90003-3

DeBruine, L. M., & Barr, D. J. (2021). Understanding Mixed-Effects Models Through Data Simulation. *Advances in Methods and Practices in Psychological Science*, *4*(1), 2515245920965119. https://doi.org/10.1177/2515245920965119

Dhar, R., & Gorlin, M. (2013). A dual-system framework to understand preference construction processes in choice. *Journal of Consumer Psychology*, *23*(4), 528–542. https://doi.org/10.1016/j.jcps.2013.02.002

Djebbara, Z., & Kalantari, S. (2023). Affordances and curvature preference: The case of real objects and spaces. *Annals of the New York Academy of Sciences*, *1527*(1), 14–19. https://doi.org/10.1111/nyas.15038

Elliot, A. J., Eder, A. B., & Harmon-Jones, E. (2013). Approach–Avoidance Motivation and Emotion: Convergence and Divergence. *Emotion Review*, *5*(3), 308–311. https://doi.org/10.1177/1754073913477517

Friedenberg, J., & Liby, B. (2016). Perceived beauty of random texture patterns: A preference for complexity. *Acta Psychologica*, *168*, 41–49. https://doi.org/10.1016/j.actpsy.2016.04.007

Gibson, J. J. (2014). *The Ecological Approach to Visual Perception: Classic Edition*. Psychology Press. https://doi.org/10.4324/9781315740218

Graf, L. K. M., & Landwehr, J. R. (2015). A Dual-Process Perspective on Fluency-Based Aesthetics: The Pleasure-Interest Model of Aesthetic Liking. *Personality and Social Psychology Review*, *19*(4), 395–410. https://doi.org/10.1177/1088868315574978

Gray, J. A. (1982). On mapping anxiety. *Behavioral and Brain Sciences*, *5*(3), 506–534. https://doi.org/10.1017/S0140525X00013297

Green, P., & MacLeod, C. J. (2016). SIMR: An R package for power analysis of generalized linear mixed models by simulation. *Methods in Ecology and Evolution*, *7*(4), 493–498. https://doi.org/10.1111/2041-210X.12504

Greenwald, A. G., Poehlman, T. A., Uhlmann, E. L., & Banaji, M. R. (2009). Understanding and using the Implicit Association Test: III. Meta-analysis of predictive validity. *Journal of Personality and Social Psychology*, *97*(1), 17–41. https://doi.org/10.1037/a0015575

Güçlütürk, Y., Jacobs, R. H. A. H., & Lier, R. van. (2016). Liking versus Complexity: Decomposing the Inverted U-curve. *Frontiers in Human Neuroscience*, *10*. https://doi.org/10.3389/fnhum.2016.00112

Havermans, R. C. (2011). "You Say it's Liking, I Say it's Wanting …". On the difficulty of disentangling food reward in man. *Appetite*, *57*(1), 286–294. https://doi.org/10.1016/j.appet.2011.05.310

Hayn-Leichsenring, G. U., & Chatterjee, A. (2018, December). *Colliding Terminological*

Systems—Immanuel Kant and Contemporary Empirical Aesthetics. https://journals.sagepub.com/doi/abs/10.1177/0276237418818635

Hofmann, W., De Houwer, J., Perugini, M., Baeyens, F., & Geert, C. (2010). Evaluative Conditioning in Humans: A Meta-Analysis. *Psychological Bulletin*, *136*(3), 390–421. https://doi.org/10.1037/a0018916

Imamoglu, Ç. (2000). Complexity, liking and familiarity: Architecture and non-architecture Turkish students' assessments of traditional and modern house facades. *Journal of Environmental Psychology*, *20*(1), 5–16. https://doi.org/10.1006/jevp.1999.0155

Iyer, E. S. (1989). Unplanned Purchasing: Knowledge Of Shopping Environment And. *Journal of Retailing; Greenwich*, *65*(1), 40.

Jacobsen, T. (2010). Beauty and the brain: Culture, history and individual differences in aesthetic appreciation. *Journal of Anatomy*, *216*(2), 184–191. https://doi.org/10.1111/j.1469-7580.2009.01164.x

Jacobsen, T., & Höfel, L. (2001). Aesthetics Electrified: An Analysis of Descriptive Symmetry and Evaluative Aesthetic Judgment Processes Using Event-Related Brain Potentials. *Empirical Studies of the Arts*, *19*(2), 177–190. https://doi.org/10.2190/P7W1-5F1F-NJK9-X05B

Jacobsen, T., & Höfel, L. (2002). Aesthetic Judgments of Novel Graphic Patterns: Analyses of Individual Judgments. *Perceptual and Motor Skills*, *95*(3), 755–766. https://doi.org/10.2466/pms.2002.95.3.755

Jacobsen, T., Schubotz, R. I., Höfel, L., & Cramon, D. Y. v. (2006). Brain correlates of aesthetic judgment of beauty. *NeuroImage*, *29*(1), 276–285. https://doi.org/10.1016/j.neuroimage.2005.07.010

Jones, A. L., & Jaeger, B. (2019). Biological Bases of Beauty Revisited: The Effect of Symmetry, Averageness, and Sexual Dimorphism on Female Facial Attractiveness. *Symmetry*, *11*(2), Article 2. https://doi.org/10.3390/sym11020279

Judd, C. M., Westfall, J., & Kenny, D. A. (2017). Experiments with More Than One Random Factor: Designs, Analytic Models, and Statistical Power. *Annual Review of Psychology*, *68*(1), 601–625. https://doi.org/10.1146/annurev-psych-122414-033702

Kahneman, D. (2011). *Thinking, Fast and Slow*. Macmillan.

Kahneman, D., Fredrickson, B. L., Schreiber, C. A., & Redelmeier, D. A. (1993). When More Pain Is Preferred to Less: Adding a Better End. *Psychological Science*, *4*(6), 401–405. https://doi.org/10.1111/j.1467-9280.1993.tb00589.x

Kakoschke, N., Hawker, C., Castine, B., de Courten, B., & Verdejo-Garcia, A. (2018). Smartphone-based cognitive bias modification training improves healthy food choice in obesity: A pilot study. *European Eating Disorders Review*, *26*(5), 526–532. https://doi.org/10.1002/erv.2622

Kumle, L., Võ, M. L.-H., & Draschkow, D. (2021). Estimating power in (generalized) linear mixed models: An open introduction and tutorial in R. *Behavior Research Methods*. https://doi.org/10.3758/s13428-021-01546-0

Leder, H., Belke, B., Oeberst, A., & Augustin, D. (2004). A model of aesthetic appreciation and aesthetic judgments. *British Journal of Psychology (London, England: 1953)*, *95*(Pt 4), 489–508. https://doi.org/10.1348/0007126042369811

Leder, H., & Nadal, M. (2014). Ten years of a model of aesthetic appreciation and aesthetic judgments: The aesthetic episode – Developments and challenges in empirical aesthetics. *British Journal of Psychology*, *105*(4), 443–464. https://doi.org/10.1111/bjop.12084

Leder, H., Tinio, P. P. L., Brieber, D., Kröner, T., Jacobsen, T., & Rosenberg, R. (2019). Symmetry Is Not a Universal Law of Beauty. *Empirical Studies of the Arts*, *37*(1), 104–114. https://doi.org/10.1177/0276237418777941

Little, A. C. (2014). Domain Specificity in Human Symmetry Preferences: Symmetry is Most Pleasant When Looking at Human Faces. *Symmetry*, *6*(2), Article 2. https://doi.org/10.3390/sym6020222

Lüdecke, D., Makowski, D., Waggoner, P., & Patil, I. (2020). *performance: Assessment of Regression Models Performance* [Computer software]. https://easystats.github.io/performance/

Luke, S. G. (2017). Evaluating significance in linear mixed-effects models in R. *Behavior Research Methods*, *49*(4), 1494–1502. https://doi.org/10.3758/s13428-016-0809-y

Marin, M. M., Lampatz, A., Wandl, M., & Leder, H. (2016). Berlyne Revisited: Evidence for the Multifaceted Nature of Hedonic Tone in the Appreciation of Paintings and Music. *Frontiers in Human Neuroscience*, *10*. https://doi.org/10.3389/fnhum.2016.00536

Markman, A. B., & Brendl, C. M. (2005). Constraining Theories of Embodied Cognition: *Psychological Science*. https://journals.sagepub.com/doi/10.1111/j.0956-7976.2005.00772.x

Martindale, C., Moore, K., & Borkum, J. (1990). Aesthetic Preference: Anomalous Findings for Berlyne's Psychobiological Theory. *The American Journal of Psychology*, *103*(1), 53–80. JSTOR. https://doi.org/10.2307/1423259

Mather, K. B., Aleem, H., Rhee, Y., & Grzywacz, N. M. (2023). Social groups and polarization of aesthetic values from symmetry and complexity. *Scientific Reports*, *13*(1), 21507.

McDougall, S. J. P., & Reppa, I. (2008). Why do I like It? The Relationships between Icon

Characteristics, user Performance and Aesthetic Appeal. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *52*(18), 1257–1261. https://doi.org/10.1177/154193120805201822

Menninghaus, W., Wagner, V., Wassiliwizky, E., Schindler, I., Hanich, J., Jacobsen, T., & Koelsch, S. (2019). What are aesthetic emotions? *Psychological Review*, *126*(2), 171–195. https://doi.org/10.1037/rev0000135

Meule, A., Richard, A., Dinic, R., & Blechert, J. (2019). Effects of a Smartphone-Based Approach-Avoidance Intervention on Chocolate Craving and Consumption: Randomized Controlled Trial. *JMIR mHealth and uHealth*, *7*(11), e12298. https://doi.org/10.2196/12298

Møller, A. P., & Thornhill, R. (1998). Bilateral Symmetry and Sexual Selection: A Meta-Analysis. *The American Naturalist*, *151*(2), 174–192. https://doi.org/10.1086/286110

Nadal, M., Munar, E., Marty, G., & Cela-Conde, C. J. (2010). Visual Complexity and Beauty Appreciation: Explaining the Divergence of Results. *Empirical Studies of the Arts*, *28*(2), 173–191. https://doi.org/10.2190/EM.28.2.d

Palmer, S. E., Schloss, K. B., & Sammartino, J. (2013). Visual Aesthetics and Human Preference. *Annual Review of Psychology*, *64*(1), 77–107. https://doi.org/10.1146/annurev-psych-120710-100504

Peirce, J., Gray, J. R., Simpson, S., MacAskill, M., Höchenberger, R., Sogo, H., Kastman, E., & Lindeløv, J. K. (2019). PsychoPy2: Experiments in behavior made easy. *Behavior Research Methods*, *51*(1), 195–203. https://doi.org/10.3758/s13428-018-01193-y

Pieters, R., & Warlop, L. (1999). Visual attention during brand choice: The impact of time pressure and task motivation. *International Journal of Research in Marketing*, *16*(1), 1–

16. https://doi.org/10.1016/S0167-8116(98)00022-6

Piqueras-Fiszman, B., Kraus, A. A., & Spence, C. (2014). “Yummy” versus “Yucky”! Explicit and implicit approach–avoidance motivations towards appealing and disgusting foods. *Appetite*, *78*, 193–202. https://doi.org/10.1016/j.appet.2014.03.029

Pombo, M., Aleem, H., & Grzywacz, N. M. (2023). Multiple Axes of Visual Symmetry: Detection and Aesthetic Preference. *Symmetry*, *15*(8), Article 8. https://doi.org/10.3390/sym15081568

Pombo, M., & Velasco, C. (2021). How aesthetic features convey the concept of brand premiumness. *Psychology & Marketing*, *38*(9), 1475–1497. https://doi.org/10.1002/mar.21534

Rangel, A., Camerer, C., & Montague, P. R. (2008). A framework for studying the neurobiology of value-based decision making. *Nature Reviews Neuroscience*, *9*(7), Article 7. https://doi.org/10.1038/nrn2357

Reber, R., Schwarz, N., & Winkielman, P. (2004). Processing Fluency and Aesthetic Pleasure: Is Beauty in the Perceiver’s Processing Experience? *Personality and Social Psychology Review*, *8*(4), 364–382. https://doi.org/10.1207/s15327957pspr0804_3

Reppa, I., & McDougall, S. (2022). Aesthetic appeal influences visual search performance. *Attention, Perception, & Psychophysics*, *84*(8), 2483–2506. https://doi.org/10.3758/s13414-022-02567-3

Reutskaja, E., Nagel, R., Camerer, C. F., & Rangel, A. (2011). Search Dynamics in Consumer Choice under Time Pressure: An Eye-Tracking Study. *American Economic Review*, *101*(2), 900–926. https://doi.org/10.1257/aer.101.2.900

Rinck, M., & Becker, E. S. (2007). Approach and avoidance in fear of spiders. *Journal of*

Behavior Therapy and Experimental Psychiatry, *38*(2), 105–120. https://doi.org/10.1016/j.jbtep.2006.10.001

Rougier, M., Muller, D., Courset, R., Smeding, A., Devos, T., & Batailler, C. (2020). Toward the use of approach/avoidance tendencies as attitude measures: Individual- and group-level variability of the ingroup bias. *European Journal of Social Psychology*, *50*(4), 857–875. https://doi.org/10.1002/ejsp.2653

Rougier, M., Muller, D., Ric, F., Alexopoulos, T., Batailler, C., Smeding, A., & Aubé, B. (2018). A new look at sensorimotor aspects in approach/avoidance tendencies: The role of visual whole-body movement information. *Journal of Experimental Social Psychology*, *76*, 42–53. https://doi.org/10.1016/j.jesp.2017.12.004

Schneider, J., & Hall, J. (2011). Why Most Product Launches Fail. *Harvard Business Review*.

Shenhav, A. (2024). The affective gradient hypothesis: An affect-centered account of motivated behavior. *Trends in Cognitive Sciences*. https://doi.org/10.1016/j.tics.2024.08.003

Silvia, P. J. (2005). *Emotional Responses to Art: From Collation and Arousal to Cognition and Emotion*. https://doi.org/10.1037/1089-2680.9.4.342

Skov, M. (2019). Aesthetic Appreciation: The View from Neuroimaging. *Empirical Studies of the Arts*, *37*(2), 220–248. https://doi.org/10.1177/0276237419839257

Skov, M., & Nadal, M. (2020). A Farewell to Art: Aesthetics as a Topic in Psychology and Neuroscience. *Perspectives on Psychological Science*, *15*(3), 630–642. https://doi.org/10.1177/1745691619897963

Solanki, J., Joshi, N., Shah, C., Hb, M., & Pa, G. (2012). A Study of Correlation between Auditory and Visual Reaction Time in Healthy Adults. *International Journal of Medicine and Public Health*, *2*(2), 36–38. https://doi.org/10.5530/ijmedph.2.2.8

Solarz, A. (1960). Latency of instrumental responses as a function of compatibility with the meaning of eliciting verbal signs. *Journal of Experimental Psychology*, *59*(4), 239–245. https://doi.org/10.1037/h0047274

Stamps, A. E. (2002). Entropy, Visual Diversity, and Preference. *The Journal of General Psychology*, *129*(3), 300–320. https://doi.org/10.1080/00221300209602100

Steciuch, C. C., Kopatich, R. D., Feller, D. P., Durik, A. M., & Millis, K. (2019). Don't go with your gut: Exploring the role of motivation in aesthetic experiences. *Psychology of Aesthetics, Creativity, and the Arts*, No Pagination Specified-No Pagination Specified. https://doi.org/10.1037/aca0000259

Tawil, N., Elias, J., Ascone, L., & Kühn, S. (2024). The curvature effect: Approach-avoidance tendencies in response to interior design stimuli. *Journal of Environmental Psychology*, *93*, 102197. https://doi.org/10.1016/j.jenvp.2023.102197

Tinio, P. P. L., & Leder, H. (2009). Just how stable are stable aesthetic features? Symmetry, complexity, and the jaws of massive familiarization. *Acta Psychologica*, *130*(3), 241–250. https://doi.org/10.1016/j.actpsy.2009.01.001

Treder, M. S. (2010). Behind the Looking-Glass: A Review on Human Symmetry Perception. *Symmetry*, *2*(3), Article 3. https://doi.org/10.3390/sym2031510

Turoman, N., Velasco, C., Chen, Y.-C., Huang, P.-C., & Spence, C. (2018). Symmetry and its role in the crossmodal correspondence between shape and taste. *Attention, Perception, & Psychophysics*, *80*(3), 738–751. https://doi.org/10.3758/s13414-017-1463-x

Vartanian, O., Navarrete, G., Chatterjee, A., Fich, L. B., Leder, H., Modroño, C., Nadal, M., Rostrup, N., & Skov, M. (2013). Impact of contour on aesthetic judgments and approach-avoidance decisions in architecture. *Proceedings of the National Academy of Sciences*,

110(Supplement 2), 10446–10453. https://doi.org/10.1073/pnas.1301227110

Vartanian, O., Navarrete, G., Chatterjee, A., Fich, L. B., Leder, H., Modroño, C., Rostrup, N., Skov, M., Corradi, G., & Nadal, M. (2019). Preference for Curvilinear Contour in Interior Architectural Spaces: Evidence from Experts and Nonexperts. *Psychology of Aesthetics, Creativity, and the Arts*, *13*(1), 110–116. https://doi.org/10.1037/aca0000150

Velasco, C., Pathak, A., Woods, A. T., Corredor, A., & Elliot, A. J. (2020). The relation between symmetry in food packaging and approach and avoidance words. *Quarterly Journal of Experimental Psychology*, *73*(5), 654–663. https://doi.org/10.1177/1747021819887172

Velasco, C., Salgado-Montejo, A., Elliot, A. J., Woods, A. T., Alvarado, J., & Spence, C. (2016). The shapes associated with approach/avoidance words. *Motivation and Emotion*, *40*(5), 689–702. https://doi.org/10.1007/s11031-016-9559-5

Vessel, E. A., Maurer, N., Denker, A. H., & Starr, G. G. (2018). Stronger shared taste for natural aesthetic domains than for artifacts of human culture. *Cognition*, *179*, 121–131. https://doi.org/10.1016/j.cognition.2018.06.009

Vessel, E. A., & Rubin, N. (2010). Beauty and the beholder: Highly individual taste for abstract, but not real-world images. *Journal of Vision*, *10*(2), 18–18. https://doi.org/10.1167/10.2.18

Wang, H.-F. (2019). "Think of the Children!": The Relationship Between Visual Complexity, Age, Visual Aesthetics, and Learning Motivation with Regard to Children. In K. Blashki & P. Isaías (Eds.), *Handbook of Research on Human-Computer Interfaces and New Modes of Interactivity:* IGI Global. https://doi.org/10.4018/978-1-5225-9069-9

Wang, L., Yu, Y., & Li, O. (2019). The typeface curvature effect: The role of typeface curvature in increasing preference toward hedonic products. *Psychology & Marketing*, *n/a*(n/a).

https://doi.org/10.1002/mar.21287

Wassiliwizky, E., & Menninghaus, W. (2021). Why and How Should Cognitive Science Care about Aesthetics? *Trends in Cognitive Sciences*, *25*(6), 437–449. https://doi.org/10.1016/j.tics.2021.03.008

Weichselbaum, H., Leder, H., & Ansorge, U. (2018). Implicit and Explicit Evaluation of Visual Symmetry as a Function of Art Expertise. *I-Perception*, *9*(2), 2041669518761464. https://doi.org/10.1177/2041669518761464

Weil, R., Feist, A., Moritz, S., & Wittekind, C. E. (2017). Approaching contamination-related stimuli with an implicit Approach-Avoidance Task: Can it reduce OCD symptoms? An online pilot study. *Journal of Behavior Therapy and Experimental Psychiatry*, *57*, 180–188. https://doi.org/10.1016/j.jbtep.2017.05.007

Wittekind, C. E., Feist, A., Schneider, B. C., Moritz, S., & Fritzsche, A. (2015). The approach-avoidance task as an online intervention in cigarette smoking: A pilot study. *Journal of Behavior Therapy and Experimental Psychiatry*, *46*, 115–120. https://doi.org/10.1016/j.jbtep.2014.08.006

Woods, A. T., Velasco, C., Levitan, C. A., Wan, X., & Spence, C. (2015). Conducting perception research over the internet: A tutorial review. *PeerJ*, *3*, e1058. https://doi.org/10.7717/peerj.1058

Zajonc, R. B. (1968). Attitudinal effects of mere exposure. *Journal of Personality and Social Psychology*, *9*(2, Pt.2), 1. https://doi.org/10.1037/h0025848

Zech, H. G., Rotteveel, M., van Dijk, W. W., & van Dillen, L. F. (2020). A mobile approach-avoidance task. *Behavior Research Methods*. https://doi.org/10.3758/s13428-020-01379-3

Supplementary Materials

S1 - Pilot Study

Methods

This study aimed to follow the same design as that proposed for Experiment 1. The stimuli, design, procedure, and data analysis are exactly the same.

Participants: 200 participants took part in our pilot study (124 female, ages 18 to 70, M = 33.73, SD = 12.95). Half of the participants were assigned to the symmetry condition and half were assigned to the complexity condition. It took participants, on average, approximately 8 minutes to complete the study. Participants were recruited through Prolific Academic (https://prolific.co) and only those who indicated that they were U.K. nationals, speak English as a first language, and have normal or corrected-to-normal vision were allowed to participate. Participants were paid according to Prolific's standards.

Results

We excluded data from participants who did not follow the instructions correctly (e.g., participants who got less than 20% correct in either the congruent or incongruent conditions). We excluded data from eleven participants (N = 189). Additionally, we excluded all trials in which participants answered incorrectly, that is, 17.3% of the trials (14.5 % of the symmetry trials and 20.3% of the complexity trials). Likewise, we excluded the trials in which the reaction time was outside two standard deviations from the mean reaction time for each of the aesthetic

conditions. For the symmetry condition we excluded trials with reaction times greater than 4.79 seconds (the lower bound is zero) and for the complexity condition we excluded trials with reaction times smaller than 0.34 seconds and greater than 2.39 seconds.

The results of our linear models indicate a significant effect of congruency on reaction times for both symmetry, $t(6424) = 2.65$, b = 0.02, $p < 0.008$, 95% CI = [0.01, 0.04], and complexity, $t(5824) = 3.58$, b = 0.02, $p < 0.001$, 95% CI = [0.01, 0.02]. For the symmetry condition, the mean reaction time in the congruent condition was 1.50 ($SD = 0.51$) while the mean reaction time in the incongruent condition was 1.59 ($SD = 0.62$). In other words, participants were, on average, faster at approaching symmetrical stimuli and avoiding asymmetrical stimuli, and slower at avoiding symmetrical stimuli and approaching asymmetrical stimuli. For the complexity condition, the mean reaction time in the congruent condition was 1.30 s ($SD = 0.27$) while the mean reaction time in the incongruent condition was 1.32 ($SD = 0.28$). In other words, participants were, on average, faster at approaching complex stimuli and avoiding simple stimuli, and slower at avoiding complex stimuli and approaching simple stimuli. Figure S1 provides a graphical representation of the data and Tables S1 and S2 display the details of the linear mixed effects models for the symmetry and complexity conditions respectively.

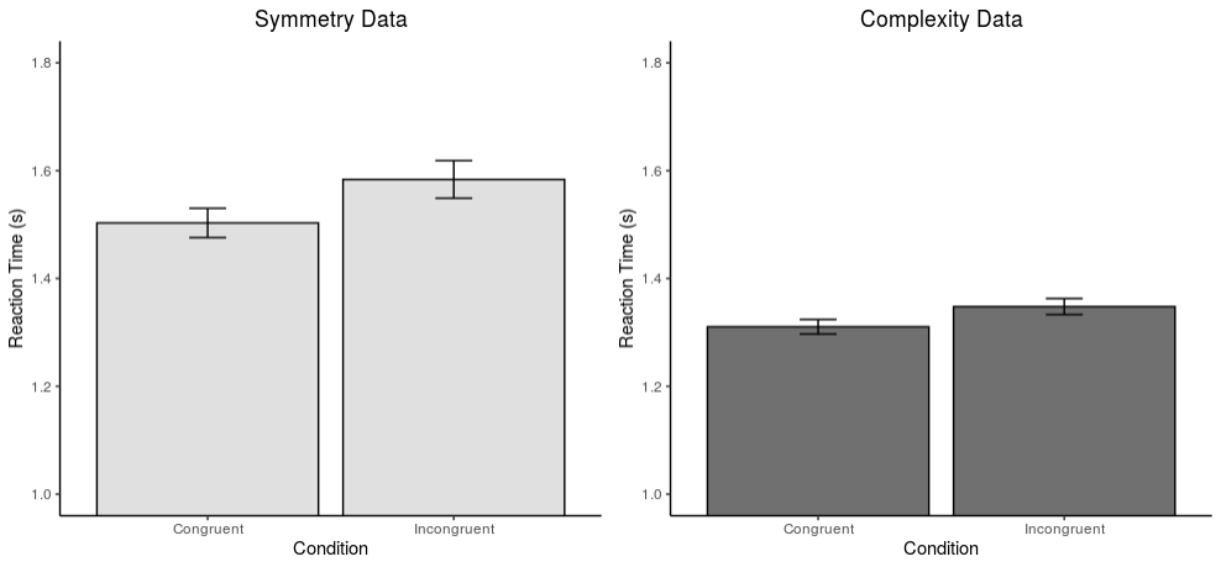


Figure S1. Bar plots of symmetry and complexity conditions, respectively. Error bars represent mean squared error.

Table S1. Mixed-effect model for symmetry condition

Random Effects

	Variance	***SD***
Stimulus	0.00	0.06
Participant	0.11	0.33

Fixed Effects

	Estimate	***SE***	**95% CI**	***df***	***t***	***p***
Intercept	0.34	0.03	[0.27, 0.41]	97.70	9.83	<0.001
Congruency (Incongruent)	0.02	0.01	[0.01, 0.04]	6228.00	2.65	0.008

This model explains 54% of the variance in log reaction times with RMSE of 0.31.

Table S2. Mixed-effect model for complexity condition

Random Effects

	Variance	***SD***
Stimulus	0.00	0.04
Participant	0.01	0.11

Fixed Effects

	Estimate	***SE***	**95% CI**	***df***	***t***	***p***
Intercept	0.25	0.01	[0.23, 0.27]	118.10	21.29	<0.001
Congruency (Incongruent)	0.02	0.00	[0.01, 0.02]	5630.00	3.58	<0.001

This model explains 31% of the variance in log reaction times with RMSE of 0.17.

Discussion

Our pilot study aimed to test the feasibility of the online VAAST in measuring the relationship between aesthetic features and approach and avoidance motivated behavior. Our results indicate a relationship between approach and avoidance motivation and two specific aesthetic features: symmetry and complexity.